Part IV

Volatility as an Asset Class

Volatility stopped being a parameter and became something traded in its own right. This part covers the models that reproduce a surface, the instruments that give direct exposure to variance, and what follows in market structure when the hedging those instruments require is large enough to move the market. Chapter 21 carries the dagger; Chapters 22 and 23 do not, and describe episodes (the February 2018 unwind, the rise of same-day options) that a reader of the financial press has already met.

21 Volatility Modeling: Local, Stochastic, Jump, and Rough
the model families, and which failure each one accepts

22 Volatility Products: VIX, Variance Swaps, and Dispersion
trading variance directly

23 Short-Dated Options, Dealer Positioning, and Market Feedback
when hedging flows move the underlying

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Chapter 21

Volatility Modeling: Local, Stochastic, Jump, and Rough †

Section 19.4 named three families of model that resolve the contradiction between a formula with one volatility and a market that quotes many, and said what each one buys and what it costs. That is enough to tell them apart; it is not enough to choose one and defend the choice. This chapter develops how each is built, how it is calibrated, and what its particular failure costs, and adds a fourth, rough volatility, which is the most recent development and only became usable after about 2015. The mathematics is deferred to [TN21]; what stays here is what each model captures, where it fails, and how one would tell.

The organizing question is not "which model is right." No model here is right. The question is which model is wrong in a way that does not matter for the trade in front of you, and answering it requires knowing precisely how each one fails.

The order is logical rather than historical (Section 21.13 gives the history). Local volatility comes first, in Section 21.2, because an exact fit to today's surface forces it on you, and Section 21.3 shows what the fit costs in dynamics; Heston's stochastic volatility, Merton's jumps and their combination in Bates follow. Section 21.7 measures all three against one fact, the $T^{-0.4}$ power law the index skew follows, and none of them reproduces it. That fact produced the fourth family; the later sections turn to the hybrid models desks run, to calibration, and to what the disagreement between models costs.

One convention holds for every worked example in the chapter: $S_0 = 100$ and $r = q = 0$. The forward then equals the spot, "at the money" means a strike of 100 at every maturity, and the numbers measure the models rather than the carry. The equations keep r and q, because the models are used with them; the examples drop them, because they would only obscure the comparison.

21.1 WHAT A VOLATILITY MODEL MUST DO

Two requirements compete, and almost every model satisfies one at the expense of the other.

Fit today's surface. A model that does not reproduce the quoted vanilla prices is arbitrageable against the market by anyone who trades the vanillas. This is a hard constraint, not a preference.

Get the dynamics right. The model must also say how the surface will move, because the value of anything path-dependent depends on that. Section 19.11 showed the point in miniature: the delta of a plain call already depends on whether the smile is sticky in strike or in moneyness, and that is a question about dynamics, not about today's prices.

Nothing in the vanilla market pins down the dynamics. Today's surface is a snapshot of the risk-neutral distribution at each maturity; infinitely many processes produce the same set of terminal distributions and behave completely differently in between. That degree of freedom is the subject of this chapter, and Section 21.11 measures what it is worth in money. Table 21.1 states the trade each of the four families makes, and it is worth reading before the sections that

develop them: no row buys both requirements, which is why the chapter ends in a choice rather than in a winner.

Table 21.1: What each family buys and what it costs.

Family	Buys	Costs
Local volatility	Fits any arbitrage-free surface exactly; stays complete	Dynamics wrong: the forward smile (quoted on a future date) flattens; the market's does not
Stochastic volatility	Realistic dynamics, stationary forward smile	Cannot fit the short end; market incomplete
Jumps	Short-dated skew a diffusion cannot produce	Skew dies too fast with maturity; unhedgeable by construction
Rough volatility	The observed skew term structure from one parameter	Not Markovian; simulation is expensive

21.2 LOCAL VOLATILITY

Let volatility be a deterministic function of price and time:

$$dS = (r - q)\,S\,dt + \sigma_{\text{loc}}(S, t)\,S\,dW \tag{21.1}$$

Dupire's result is that for any arbitrage-free surface of call prices $C(K, T)$ there is exactly one function σ_{loc} in (21.1) reproducing every one of them, and it can be read off directly, evaluated at $S = K$, $t = T$:

$$\sigma^2_{\text{loc}}(K, T) = \frac{\dfrac{\partial C}{\partial T} + (r - q)K\dfrac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2\dfrac{\partial^2 C}{\partial K^2}} \tag{21.2}$$

Three things about (21.2) are easy to read past.

The denominator is the implied density. It is $\frac{1}{2}K^2e^{-rT}$ times the Breeden–Litzenberger density of (19.1). So local volatility is a ratio of two things the surface already contains, and it exists precisely when the density is positive and the numerator is not negative, which are Section 19.5's butterfly and calendar conditions again, now as an existence condition for a model.

The model is complete. There is one Brownian motion and one traded asset, so Section 13.4's replication argument survives intact and every option is hedgeable with the underlying. That is a real advantage: a local-volatility book has a delta that means something.

The fit is exact by construction, not by calibration. There are no parameters to search over. This is why local volatility remains the baseline for any payoff fixed by today's surface alone, such as an unquoted European.

The simplest local volatility is a single power of the price, the constant elasticity of variance model of Cox (1975), $\sigma_{\text{loc}}(S) = \sigma S^{\beta-1}$ with σ a scale constant: with $\beta < 1$ volatility rises as the price falls, a skew from one parameter, and $\beta = 1$ and $\beta = 0$ are the lognormal and normal cases. It fits no particular surface, but its exponent survives as SABR's β (Section 29.9).

Example 21.1 Local volatility is about twice as steep as implied

Take the surface Heston's model (Section 21.4) gives at the base case of Figure 21.2 (any arbitrage-free surface will do) and apply (21.2). Its call prices come from Heston's semi-analytic formula, and the strike derivatives are central differences in log-strike $y = \log K$ (with $C_y = \partial C/\partial y$, $K^2\,\partial^2 C/\partial K^2 = C_{yy} - C_y$), which keeps the wings numerically stable, and $\partial C/\partial T$ is a central

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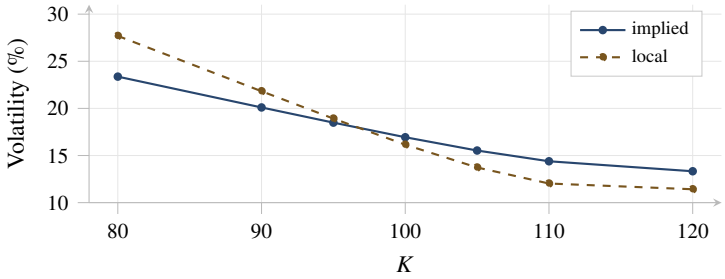


Figure 21.1: Local and implied volatility at one year, from the same Heston surface.

difference in maturity. These are formula prices, not quotes, so the noise that differencing real quotes produces ([TN21]) does not arise here. Both functions are read at the same argument K, but it means different things on the two sides: for σ_{impl} it is the strike of an option expiring in a year, and for σ_{loc} it is the price level at which that volatility is in force, a year from now. With the chapter's convention, $S_0 = 100$ and $r = q = 0$:

Strike	Implied vol	Local vol
80	23.37%	27.69%
90	20.10%	21.82%
95	18.49%	18.92%
100	16.94%	16.15%
105	15.53%	13.73%
110	14.39%	12.03%
120	13.33%	11.42%

Measured as slopes in $k = \log(K/S_0)$ at the money, across the ±5% pair (the convention (21.3) fixes), implied volatility falls at -0.296 per unit of k (about 3 volatility points from 95 to 105) and local volatility at -0.518, a ratio of **1.75**.

Figure 21.1 shows the two curves together. This is the "rule of two," and the reason for it is an averaging argument. An implied volatility is the single number that reproduces the price of an option struck at K, so it is an average of the local volatilities the price path visits on its way to K. Averaging flattens. To produce a given slope in the average, the underlying function must slope about twice as steeply.

So a steep local-volatility surface, however alarming its wings look, is not a calibration failure; it is arithmetic.

21.3 WHAT LOCAL VOLATILITY GETS WRONG

Section 19.4 said local volatility fits today's prices exactly and forecasts their evolution badly. Here is the mechanism and its size.

Because σ_{loc} is a fixed function of S and t, the model has no freedom left once it is calibrated. In particular it has a definite prediction about the **forward smile**: the smile that will be quoted at some future date, for options starting then. That prediction is wrong in a specific direction.

One measurement convention is needed first, because from here on the chapter reports skews as numbers rather than shapes. Section 19.1 used "skew" for the shape: implied volatility falling as the strike rises. Measured, it is the slope of implied volatility in **log-moneyness**, taken at the money, and the tables below report the second derivative beside it:

$$\text{skew}(T) = \left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}, \qquad \text{curvature}(T) = \left.\frac{\partial^2 \sigma_{\text{impl}}}{\partial k^2}\right|_{k=0} \tag{21.3}$$

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with $k = \log(K/F)$, where F is the forward price to the option's maturity, $F = S_0 e^{(r-q)T}$ for an asset paying a continuous dividend yield q. Under the chapter's convention $r = q = 0$, so $F = S_0$ throughout and "at the money" and "at the forward" name the same strike. **Curvature** is the smile's convexity in the same variable: zero for a straight line, and positive when both wings sit above the tangent at the money. Both are estimated in practice by finite differences across a pair of strikes either side of the forward (and the forward itself, for the curvature): placed symmetrically in k, at Fe^{-h} and Fe^{h}, where a table states a bump h, and at 95 and 105 (the "±5% strike pair") where it does not.

Three properties are worth fixing now, because every table in this chapter reports them. *Log-moneyness rather than strike* makes the number scale-free, so two underlyings trading at 40 and 4,000 report the same skew for the same shape. *The sign is negative on an equity surface*, and a steeper skew is a more negative number. *The value depends on the width of the strike pair*, because the smile's slope changes away from the money: the one-year base-case smile of Section 21.4 measures −0.296 across the ±5% strike pair and −0.282 at $h = 0.10$. Each table states its own bump, and skews are comparable down a column rather than across tables.

Example 21.2 The forward smile flattens

Take Example 21.1's surface and measure both skews ±5% around the forward, the local one at the date equal to each maturity:

Maturity	Implied skew	Local-vol skew
3 months	−0.479	−0.981
6 months	−0.403	−0.770
1 year	−0.296	−0.518
2 years	−0.196	−0.320
3 years	−0.148	−0.231

The local volatility function's own skew decays by a factor of more than four between three months and three years. A one-year option starting in three years is priced from the local volatilities between years three and four, whose skew starts at −0.231, where today's one-year option is priced from skews of −0.98 to −0.52; so the model predicts that the three-year-forward smile will be far flatter than today's spot smile.

The market does not agree. Forward-starting options (options whose strike is set at the money on a future date) and cliquets (chains of them that reset one after another; Section 34.12) trade with a skew comparable to today's: the smile is roughly *stationary* in moneyness, not decaying. So a local-volatility model marks too cheap anything that gains when the future skew steepens (an out-of-the-money forward-starting put, a forward-starting risk reversal that is long the put and short the call), marks its mirror image too expensive, and, worse, hedges both with the wrong vega profile.

Two products with identical vanilla hedges can have quite different forward-smile exposure, and local volatility, holding one flattened forward smile, cannot tell them apart.

There is a second, subtler consequence. Because local volatility ties σ to S, a fall in the spot mechanically raises volatility along the calibrated function. That makes the model's delta systematically different from the Black–Scholes delta, in a direction Section 19.11's vocabulary can name: a rise in the spot *raises* a fixed strike's implied volatility under sticky delta and *lowers* it under local volatility, so the local-volatility delta sits below even the sticky-strike (Black–Scholes) delta. The measured index behavior of Section 19.11 lies between Black–Scholes and local volatility: below the first, but not as far below as the second puts it.

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21.4 STOCHASTIC VOLATILITY

Let volatility be a second random process, correlated with the price:

$$dS = (r-q)S\,dt + \sqrt{v}\,S\,dW_1, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2, \qquad dW_1\,dW_2 = \rho\,dt \quad (21.4)$$

The variance v mean-reverts at rate κ toward a long-run level θ, with volatility-of-volatility σ_v, and it is correlated with the price at ρ. The square root in the diffusion keeps v nonnegative, though it can touch zero unless $2\kappa\theta \geq \sigma_v^2$, the Feller condition, which fitted equity parameters usually violate (the base case used throughout the chapter, $v_0 = \theta = 0.04$, $\kappa = 1.5$, $\sigma_v = 0.6$, $\rho = -0.7$, has $2\kappa\theta = 0.12$ against $\sigma_v^2 = 0.36$); that matters for simulation, where a discretized variance can go negative, not for the formula. The square root is also what makes the model tractable: the characteristic function $\varphi(u) = \mathbb{E}\left[e^{iu\log S_T}\right]$ (risk-neutral, $i = \sqrt{-1}$) is available in closed form, and a call price follows from it by Fourier inversion rather than by simulation,

$$C = S_0 e^{-qT} P_1 - K e^{-rT} P_2, \qquad P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\varphi_j(u)}{iu}\right] du \quad (21.5)$$

where $\varphi_2 = \varphi$ and $\varphi_1(u) = \varphi(u - i)/\varphi(-i)$, which is the same function under the measure that takes the stock itself as numeraire; P_2 is the risk-neutral probability that the call finishes in the money and P_1 the same probability under that stock measure, Heston's $N(d_2)$ and $N(d_1)$. And φ is elementary. Writing $b = \kappa - \rho\sigma_v iu$ and

$$d = \sqrt{b^2 + \sigma_v^2\left(u^2 + iu\right)}, \qquad g = \frac{b-d}{b+d}$$

the characteristic function is

$$\varphi(u) = \exp\left\{iu\left(\log S_0 + (r-q)T\right) + A(T) + B(T)\,v_0\right\}$$

with

$$B(T) = \frac{b-d}{\sigma_v^2}\cdot\frac{1-e^{-dT}}{1-ge^{-dT}}, \qquad A(T) = \frac{\kappa\theta}{\sigma_v^2}\left[(b-d)T - 2\log\frac{1-ge^{-dT}}{1-g}\right]$$

That is the whole of it: a square root, an exponential and a logarithm in u, with no integral and no recursion anywhere. This is what "semi-closed form" means: no formula for the price, but none needed, because everything except a single one-dimensional integral is written above. It is also why a whole surface calibrates in seconds. One warning: d and the logarithm are complex, and the algebra pins only d^2, which either square root satisfies, and the two give algebraically equivalent formulas for A and B. Take the principal root, as written above, and the logarithm stays continuous; take the other and it jumps by $2\pi i$, destroying the pricing integral at long maturities. [TN21] derives all of it and calls that failure the "little Heston trap."

Heston is the standard model for three reasons that have little to do with realism. The semi-closed form is the first. The second is its five parameters (v_0, θ, κ, σ_v, ρ), enough to fit a smile and few enough to interpret. The third is that every parameter maps onto a feature of the surface a trader already thinks in terms of.

Figure 21.2 shows the two parameters a trader adjusts most. Table 21.2 puts numbers on four of the five; the fifth, v_0, stays at 0.04 throughout. The table's skew and curvature are first and second central differences of implied volatility in log-moneyness at $h = 0.10$.

Correlation produces the skew, and produces it for the right reason. At $\rho = 0$ the skew vanishes: the smile is symmetric to three decimals, which is Section 19.1's currency shape. Making ρ negative tilts it, and the economics is Section 19.9's leverage explanation written as a parameter: when volatility rises as the price falls, low strikes become expensive.

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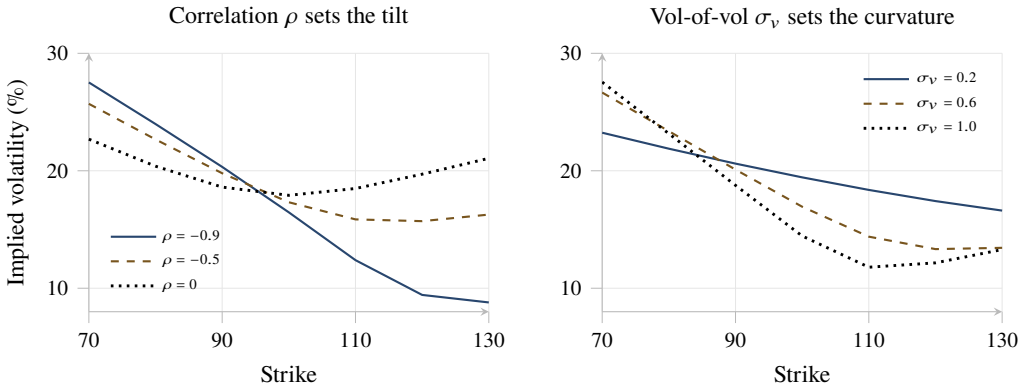


Figure 21.2: One-year Heston smiles. Base case $v_0 = \theta = 0.04$, $\kappa = 1.5$, $\sigma_v = 0.6$, $\rho = -0.7$; each panel varies one parameter.

Table 21.2: One-year at-the-money level, skew, and curvature as each parameter is varied from the base case.

	ATM (%)	Skew	Curvature
Base ($\rho = -0.7$, $\sigma_v = 0.6$, $\kappa = 1.5$, $\theta = 0.04$)	16.94	−0.282	0.36
$\rho = -0.9$	16.46	−0.398	−0.58
$\rho = 0$	17.91	0.000	1.27
$\sigma_v = 0.2$	19.44	−0.112	−0.01
$\sigma_v = 1.0$	14.45	−0.340	1.40
$\kappa = 5$	18.94	−0.155	0.03
$\theta = 0.09$	22.70	−0.227	0.08

Vol-of-vol produces the curvature. At $\sigma_v = 0.2$ the curvature is −0.01 and the smile is a straight line to two decimals; at $\sigma_v = 1.0$ it is 1.40 and both wings lift. This is the mechanism of Section 19.9: uncertainty about volatility fattens both tails of the terminal distribution.

The two interact, and that is the practical difficulty. Raising σ_v from 0.2 to 1.0 also steepens the skew from −0.112 to −0.340 and drops the at-the-money level from 19.44% to 14.45%; moving ρ from 0 to −0.9 takes the curvature from 1.27 to −0.58, because the same vol-of-vol goes into tilt rather than both wings. No parameter moves one feature alone, so calibration is a genuine optimization rather than a sequence of one-dimensional fits, and the fitted parameters are correlated enough that reporting them individually is misleading.

Mean reversion flattens everything. At $\kappa = 5$ the skew falls to −0.155 and the curvature to 0.03: fast mean reversion gives variance less time to wander, so the terminal distribution is closer to lognormal. This is the averaging that flattens the skew with maturity beyond Section 19.1's change of unit, and it is why κ is hard to identify from a single maturity's smile: its effect looks like a longer maturity.

Diffusive variance fails at the short end. Variance is a diffusion, so over a vanishingly short horizon it has not had time to move, and however the parameters are set the at-the-money skew approaches a finite limit as $T \to 0$ instead of steepening as the market's does. In the base case the one-month skew is 1.77 times the one-year skew (slopes at $h = 0.01$), where the market's power law gives about 2.7. Section 21.7 measures the gap; Sections 21.5 and 21.6 are the classical alternatives, jumps alone and jumps added to Heston. Rate markets use SABR instead, built for refitting speed: its volatility is stochastic too, but an asymptotic expansion gives the implied volatility as a formula, so a whole grid of swaption smiles can be refitted many times a day (Section 29.9).

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21.5 JUMPS

Allow the price to move discontinuously. In Merton's model a diffusion is punctuated by jumps arriving at rate λ, each multiplying the price by a lognormal factor Y:

$$\frac{dS}{S} = (r - q - \lambda\bar{k})\,dt + \sigma\,dW + (Y-1)\,dN, \qquad \bar{k} = \mathbb{E}[Y] - 1 \tag{21.6}$$

where N is a Poisson process of intensity λ (a count of jumps that rises by one at random moments, λ times a year on average) and $\log Y$ is normal with mean $\mu_J = \log(1+\bar{k}) - \delta^2/2$ and variance δ^2. The $-\lambda\bar{k}$ in the drift is a compensator: it subtracts the jumps' expected contribution, so that jumps negative on average do not drag the price below the forward and $e^{-(r-q)t}S_t$ stays a martingale.

Conditional on exactly n jumps the terminal price is lognormal again, so the option is a Poisson-weighted sum of Black–Scholes–Merton values, which is what makes the model closed form:

$$C = \sum_{n=0}^{\infty} \frac{e^{-\lambda^* T}(\lambda^* T)^n}{n!}\, C_{\mathrm{BSM}}(\sigma_n, r_n)\,, \qquad \lambda^* = \lambda(1+\bar{k})$$

with $\sigma_n^2 = \sigma^2 + n\delta^2/T$ and $r_n = r - \lambda\bar{k} + n\log(1+\bar{k})/T$. Each term is the price on the paths that happen to take n jumps; the intensity is λ^* rather than λ because the sum discounts at r while each term discounts at its own r_n.

Merton's jumps are few and large, about one every two years at the $\lambda = 0.5$ used below. The variance-gamma model of Madan, Carr, and Chang (1998) and its generalization by Carr, Geman, Madan, and Yor (2002), known as CGMY, go to the other extreme: infinitely many small jumps and typically no diffusion. They fit short-dated skew well and price by the same Fourier inversion as Heston, but their skew flattens with maturity just as Merton's does, and they are used mainly to fit and interpolate vanilla smiles. The reason to reach for jumps is entirely about short maturities.

Example 21.3 What a jump does that a diffusion cannot

Take Merton's model with $\sigma = 17\%$, jump intensity $\lambda = 0.5$ a year, mean jump $\bar{k} = -10\%$, jump volatility $\delta = 13\%$:

Strike	1 month	3 months	1 year
85	30.99%	23.82%	21.21%
90	25.07%	21.78%	20.81%
95	20.57%	20.35%	20.47%
100	18.77%	19.49%	20.18%
105	18.25%	18.99%	19.94%
110	18.42%	18.73%	19.74%

At one month the smile falls from 31.0% at 85 to a low of 18.2% at 105 and turns up again beyond it, a range of 12.7 volatility points. At one year it falls all the way across the grid, from 21.2% at 85 to 19.7% at 110, a range of 1.5 points.

The smile's range collapses by a factor of more than eight as maturity goes from one month to one year. A diffusion cannot do this, because over a short horizon a continuous path cannot travel far enough to make a 15%-out-of-the-money put worth anything; only a discontinuity can.

That collapse is also the criticism. Measured as the at-the-money skew (slopes at $h = 0.01$), the jump model's ratio between one month and one year is **3.7**; the power law index skews follow (Section 21.7) gives about **2.7**. Jumps overshoot the short end by more than a third, where a diffusion undershoots, and a model fitted to one-month options will misprice one-year options unless the jump parameters are allowed to vary with maturity—which is to say, unless the model is

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abandoned.

Section 19.4 noted that jumps break completeness for a sharper reason than stochastic volatility does: a jump of unknown size cannot be hedged by any position established before it. Stochastic volatility leaves a risk no position in the stock alone can hedge, but a second option hedges it, canceling the vega term of Chapter 18's expansion, equation (18.1), as Example 18.7 did; jump risk is not hedged by the stock and any finite set of options, because no such set pays off correctly for every possible jump size. A jump is a move too large for that expansion to describe, so canceling its terms, however completely, does not reach it. Section 13.4's argument said an incomplete market has no unique price; with jumps, it has no perfect hedge either: options can reduce the exposure, but no finite set of them removes it.

Who is paid for jump risk. Because the risk cannot be hedged away, someone must hold it, and they are paid to: part of Section 19.10's volatility risk premium compensates for diffusive variance a delta hedger cannot capture perfectly, and part for a discontinuity no hedge covers at all. The second part sits in the wings and at short maturities, where Example 21.3 put the jump model's contribution. That location is convenient for calibration and treacherous for inference: a steep short-dated put wing is consistent both with a high real-world chance of jumps and with a large premium for bearing them, and option prices, which are risk-neutral, cannot separate the two.

For the desk, the consequence is a rule about who should be short. A delta-hedged book can hold diffusive variance risk and earn the premium on it. Jump risk does not diversify across stocks, because jumps cluster across underlyings (Chapter 23 shows the same clustering intraday), so a book that has systematically sold short-dated out-of-the-money options holds a correlated tail, and Chapter 22's case study of February 5, 2018 shows a crowd of short-volatility positions meeting such a tail.

21.6 STOCHASTIC VOLATILITY WITH JUMPS

One mechanism undershoots the short end and the other overshoots it, so the obvious move is to run both at once. Bates (1996) does exactly that: Heston's variance process, with Merton's jumps added to the price:

$$\frac{dS}{S} = (r - q - \lambda\bar{k})\,dt + \sqrt{v}\,dW_1 + (Y-1)\,dN, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2 \quad (21.7)$$

with $dW_1\,dW_2 = \rho\,dt$, and N, Y and $\bar{k}$ as in (21.6). The jumps are taken independent of both Brownian motions.

That independence is what makes the model usable: $\log S_T$ is a sum of independent pieces, so its characteristic function is a product:

$$\varphi_{\text{Bates}}(u) = \varphi_{\text{Heston}}(u) \cdot \exp\left\{\lambda T\left(e^{\,iu\mu_J - u^2\delta^2/2} - 1\right) - iu\lambda\bar{k}T\right\}, \qquad \mu_J = \log(1+\bar{k}) - \frac{\delta^2}{2}$$

with φ_{Heston} exactly as Section 21.4 writes it. The price follows from the same inversion (21.5), subject to the same warning about the branch of d. One integral still prices it, which is why Bates is widely used for short-dated skew; [TN21] derives the product and recovers Merton's series from it.

Three parameters are added to Heston's five, and they act where Heston cannot reach:

- λ, the arrival rate, sets how much of the smile is made by jumps rather than by the diffusion.
- $\bar{k}$, the mean jump, tilts the short-dated smile, and a negative value is what makes short-dated puts expensive.

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- δ, the jump volatility, spreads the short-dated wings.

Example 21.4 shows the division of labor in the smiles themselves.

Example 21.4 Where each mechanism acts

Take Heston's base case with its diffusive variance cut to $v_0 = \theta = 0.025$ and Example 21.3's jumps added ($\lambda = 0.5$, mean jump -10%, jump volatility 13%), so that the two mechanisms together carry the base case's variance rate of 0.04 a year:

Strike	1 month	3 months	1 year
85	31.57%	26.08%	21.67%
90	26.52%	23.58%	20.36%
95	21.67%	20.68%	19.00%
100	17.17%	17.33%	17.55%
105	13.74%	14.08%	16.03%
110	15.06%	13.21%	14.57%

At one month the smile falls from 31.6% at 85 to 13.7% at 105, a range of 17.8 volatility points, and turns up again beyond it. Over the same strikes Heston's own one-month smile spans 11.1 points and the jump model's 12.7: the combination is steeper than either component, because the jump wing now sits on a diffusion that is itself skewed. At one year the range is 7.1 points, against Heston's 7.3 and the jump model's 1.5.

At one month the shape of the smile (the put wing and the upturn beyond 105) is the jump model's; at one year the smile is Heston's, tilt and all. Each mechanism carries the maturities the other cannot reach.

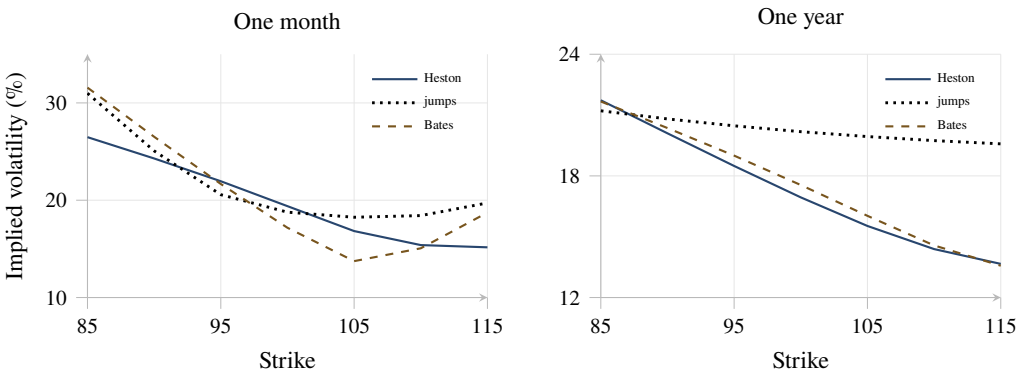


Figure 21.3: One-month and one-year smiles under Heston, Merton's jumps and Bates, with the parameters of Examples 21.3 and 21.4.

Figure 21.3 draws the comparison. In the left panel the Bates curve tracks the jump curve down the put wing and turns up with it past 105, sitting below it from the money upward because its diffusive variance is smaller, 0.025 against the jump model's $0.17^2 = 0.029$; in the right panel it lies on top of Heston's, and the jump curve is nearly flat. Table 21.3 puts numbers on what each of the three new parameters does to that picture. Every row resets $v_0 = \theta$ to hold the variance rate at the base case's; skew and curvature are central differences at $h = 0.10$, and the last column divides the one-month by the one-year slope at the money ($h = 0.01$, as in Figure 21.4), so it is not the quotient of the skew columns.

Read the one-year columns first. Down every row the skew stays between -0.24 and -0.36 and the curvature within about 1.4 of zero, small beside the one-month column's 1.3 to 19.2, so a year out the three jump parameters are close to invisible and the smile is the one Table 21.2

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Table 21.3: One-month and one-year at-the-money level, skew, and curvature as each jump parameter is varied from the Bates base case ($\lambda = 0.5$, $\bar{k} = -0.10$, $\delta = 0.13$).

		One month			One year			
	$v_0 = \theta$	ATM (%)	Skew	Curv.	ATM (%)	Skew	Curv.	**Ratio**
Base	0.0250	17.17	−0.531	7.11	17.55	−0.289	−0.44	2.92
$\lambda = 0.25$	0.0325	18.35	−0.495	3.42	17.23	−0.287	−0.02	2.23
$\lambda = 1.0$	0.0101	13.96	−0.584	17.02	18.19	−0.287	−1.28	6.20
$\bar{k} = -0.05$	0.0297	18.07	−0.380	6.34	17.35	−0.253	0.17	2.55
$\bar{k} = -0.15$	0.0169	15.19	−0.703	10.33	17.75	−0.346	−1.36	3.63
$\delta = 0.05$	0.0330	18.92	−0.538	1.29	17.49	−0.246	0.01	2.77
$\delta = 0.20$	0.0121	13.13	−0.400	19.15	16.64	−0.359	−0.13	3.18

described. At one month they are everything.

The intensity sets how much of the smile the jumps make. Doubling λ to 1.0 leaves only a quarter of the variance in the diffusion, and the one-month at-the-money level falls from 17.2% to 14.0% while the curvature rises from 7.1 to 17.0: variance that used to sit under the whole distribution now sits in its tails. The one-year row barely notices.

The mean jump sets the tilt. From $\bar{k} = -0.05$ to -0.15 the one-month skew steepens from −0.380 to −0.703, and the one-month/one-year ratio goes from 2.55 to 3.63. This is the parameter that makes short-dated puts expensive, and it is the one a calibration to the put wing pins down first.

The jump volatility sets the wings. From $\delta = 0.05$ to 0.20 the one-month curvature goes from 1.3 to 19.2, and at the top of that range the skew flattens to −0.400, because a wide jump distribution fattens both tails rather than the left one alone.

What the combination buys is the last column. The base case's one-month slope at the money is **2.92** times its one-year slope, against the market's 2.70, where Heston alone reaches 1.77 and jumps alone 3.70; Section 21.7 draws all four against maturity. Mixing two errors of opposite sign cancels most of the short-end miss.

The long end is worse, not better. At three years the skew is 0.372 of its one-year value, against the market's 0.644, almost exactly the jump model's 0.367, and further off than Heston's own 0.494. Moving variance out of the diffusion and into jumps does not lift the term structure; it rotates it: jump skew is concentrated at short maturities and is negligible at long ones, so the short-end repair is paid for at the long end. Lowering the intensity trades one end against the other (at $\lambda = 0.25$ the three-year ratio improves to 0.42 while the one-month falls to 2.23, below the market); raising it misses both ends by more (6.20 and 0.317 at $\lambda = 1.0$).

Eight parameters are enough to put the fit where a particular book needs it, which is why many equity desks use some version of the model for short-dated vanillas. They do not supply the power law the market's skew follows across maturities (Section 21.7): two unrelated mechanisms, each fitted to a different part of the surface, are a patch rather than an explanation.

21.7 THE SKEW TERM STRUCTURE, AND WHY IT BROKE ALL THREE MODELS

Sections 21.4, 21.5 and 21.6 each made a claim about how the skew behaves as maturity shortens. Put the three side by side against the market and the claims resolve into a single, sharp, measurable failure, the one the next section finally explains.

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Write $\psi(T)$ for the *magnitude* of the at-the-money skew (21.3),

$$\psi(T) = -\left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}$$

which is positive on an equity surface, so that a steepening skew is a rising ψ. Ask how it behaves as $T \to 0$.

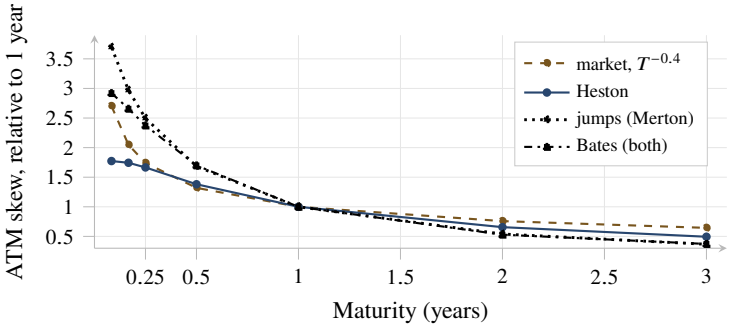


Figure 21.4: At-the-money skew against maturity, normalized to one year: Heston's base case, Example 21.3's jumps, Example 21.4's Bates, and the power law $T^{-0.4}$; slopes at the money, $h = 0.01$.

The empirical answer is a power law. Across equity indices, $\psi(T)$ is roughly proportional to $T^{-\alpha}$ with α around 0.4 (Bayer, Friz, and Gatheral 2016), at least from about a month to a few years: the skew keeps steepening as maturity shortens. Whether it goes on steepening below a month is disputed, and Section 21.8 returns to it.

A classical stochastic-volatility model cannot produce that. Section 21.4 said why: variance is a diffusion, so $\psi(T)$ approaches a finite constant as $T \to 0$ no matter how the parameters are set. In Figure 21.4 the Heston skew rises only 1.77 times from one year to one month, where the market rises 2.70 times.

At the long end the failure reverses. Heston's skew decays to 0.494 of its one-year value by three years, against the market's 0.644: too flat at the short end and decaying too fast at the long end simultaneously, which is not a parameter problem but a structural one. Fitting Heston to a whole surface therefore means fitting it badly everywhere rather than well somewhere.

Jumps fix the short end and overshoot it, as Example 21.3 showed: ratio 3.7 against the market's 2.7, the third curve in Figure 21.4. Bates, the fourth curve, splits the difference at the short end (2.92 against 2.70) and inherits the jump model's long end, 0.372 against 0.644, for the reason Section 21.6 gave.

None of the three produces the power law. In each of them the variance is either constant or a Markov diffusion, with or without jumps appended to the price, and the short-end skew of any such model either saturates or is set by hand, maturity by maturity, through the jump parameters. The market's curve is not a compromise between the other three; it is a different shape, and matching it needs either a different kind of process or several ordinary ones with widely separated speeds. That was the position for two decades: a short-maturity problem, patched and never explained.

21.8 ROUGH VOLATILITY

In a paper circulated in 2014 and published in 2018, Gatheral, Jaisson, and Rosenbaum estimated the smoothness of realized volatility directly from high-frequency data on equity indices and liquid futures, and found a pattern that had been in the data all along. Log-volatility

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behaves like the fractional Brownian motion of Section 14.13, with Hurst parameter $H \approx 0.1$. Ordinary Brownian motion has $H = 0.5$; at 0.1 the path is far *rougher* (more jagged, with successive increments correlated at −0.43 rather than zero), and the rough path of Figure 14.5 is what such a series looks like. The estimate was strikingly stable: roughly the same H, between 0.06 and 0.20 and nowhere near 0.5, across twenty-one equity indices and for Bund, crude oil and gold futures; the same when the sample was split in two; and, in later work (Bennedsen, Lunde, and Pakkanen 2022), across close to two thousand individual US stocks. Two things made this more than an empirical curiosity.

It produces the observed skew term structure with no extra machinery. A model whose volatility is driven by a fractional Brownian motion with Hurst parameter H generates

$$\psi(T) \sim T^{H-1/2} \quad \text{as } T \to 0 \tag{21.8}$$

where $\sim$ means proportional to, in that limit. Setting $H = 0.1$ gives $\psi(T) \sim T^{-0.4}$—the observed power law, from one parameter, with $H = 0.5$ recovering the flat short-end behavior of the classical models. Figure 21.4's "market" curve *is* equation (21.8) at $H = 0.1$.

And roughly the same H appears on both sides. The value estimated from realized volatility under the physical measure (the real-world probabilities that generated the data) roughly matches the value implied by option prices under the risk-neutral measure. A parameter estimated from time-series data has no obligation to match one estimated from cross-sectional prices, and when it does, that is evidence about the process rather than about the fit.

Case Study 21.1 How roughness became usable

Rough volatility was, at first, a model no one could use. Fractional Brownian motion is not Markovian: the expected future path of volatility depends on its entire history, so there is no state variable to put on a tree and no partial differential equation to solve. Every price required a simulation, and simulating the model's volatility exactly, jointly with the price, costs $O(n^2)$ per path in the number of time steps after a Cholesky factorization, against $O(n)$ for an ordinary diffusion. Three developments changed that between roughly 2015 and 2021.

Fast simulation. Hybrid schemes reduced the cost of generating rough paths to something comparable with ordinary Monte Carlo, by computing exactly the heavily weighted most recent shocks and approximating the weight on older ones.

Markovian approximations. The rough weighting of past shocks can be approximated by a sum of a small number of ordinary mean-reverting factors with widely separated speeds. This restores a state vector, typically of four to eight dimensions, and with it ordinary Monte Carlo and, in low dimension, the partial differential equations of Chapter 20. It is an approximation, but a controlled one, and it is how most implementations work.

Neural network calibration. Calibrating a rough model by repeated simulation is prohibitive. Training a network offline to map parameters to implied volatility surfaces, then inverting it at run time, turns a calibration that took hours into one that takes milliseconds. The network forecasts nothing; it interpolates a known function that is merely expensive to evaluate, so it replaces repeated simulation without changing the model being calibrated.

Adoption has been uneven. Non-bank market makers and some hedge funds took the models up first, for short-dated index options and relative-value trades between SPX and VIX options; banks have been slower, and most exotics books remain on the local-stochastic volatility of Section 21.9.

The lesson is about what a modeling constraint really is. For twenty years the profession assumed volatility must be modeled as a Markov diffusion, and most failures

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at the short end were patched by adding jumps. The Markov assumption was never an economic claim; it was a computational one, and when the computation became affordable the assumption went. It is worth asking which of today's modeling conventions are economics and which are just arithmetic we have not yet made cheap enough.

The standard implementation is the **rough Bergomi** model of Bayer, Friz, and Gatheral (2016), and it is worth writing out, because each of its three parameters has one main job:

$$\begin{aligned} \frac{dS}{S} &= (r-q)\,dt + \sqrt{v_t}\,dW_{1,t}, \qquad dW_1\,dW_2 = \rho\,dt \\ v_t &= \xi_0(t)\exp\left\{\eta\sqrt{2H}\int_0^t (t-s)^{H-1/2}\,dW_{2,s} - \frac{\eta^2}{2}\,t^{2H}\right\} \end{aligned} \tag{21.9}$$

Read it from the outside in. $\xi_0(t)$ is the **forward variance curve**, today's price of the variance to be realized at t, which follows from the variance swap rates Section 19.3 replicated from the vanilla surface and which Chapter 22 trades. It is an input rather than a parameter: the exponential has mean one, so $\mathbb{E}[v_t] = \xi_0(t)$ and the curve is reproduced exactly whatever the other three do. The integral is a version of Section 14.13's fractional Brownian motion (the Riemann–Liouville form: the same roughness H, but increments that are not stationary), written as a weighted sum of past shocks: each $dW_{2,s}$ enters with weight $(t-s)^{H-1/2}$, the **kernel** of Section 14.13 (Heston's is $e^{-\kappa(t-s)}$), and the $\sqrt{2H}$ normalizes the variance to t^{2H}, which is why $-\frac{1}{2}\eta^2 t^{2H}$ keeps the mean of the exponential at one. η scales the volatility of log variance and ρ the correlation, with the same jobs as Heston's σ_v and ρ.

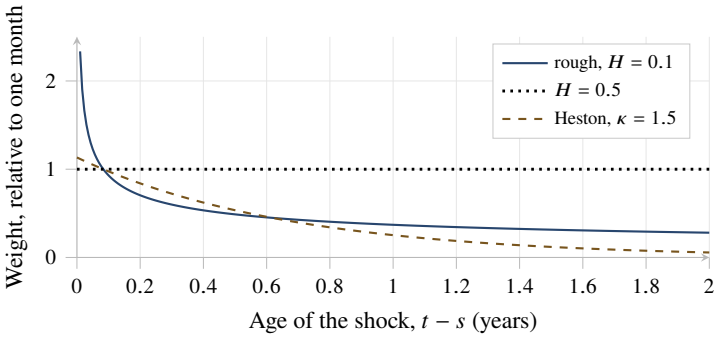


Figure 21.5: Weight on a past variance shock as a function of its age, normalized to one at one month.

The kernel is where the model differs from everything before it, and Figure 21.5 draws it against Heston's. For $H < 1/2$ the exponent is negative and the weight rises without limit as $s \to t$: the most recent shocks dominate, which is roughness. But the weight decays as a power rather than an exponential, so a shock from two years ago still carries 28% of the weight of one from a month ago, where Heston's $e^{-\kappa(t-s)}$ at $\kappa = 1.5$ leaves it 6%. Rough volatility is both more jagged and longer-remembering than a mean-reverting diffusion, and the combination is what a single exponential cannot produce: to move that much over short horizons, Heston needs a very large σ_v; to remember that long, a very small κ; and the two settings fight over the same one-year smile. Table 21.4 puts numbers on the three parameters in the format of Tables 21.2 and 21.3. Its figures are simulated (200,000 paths, hybrid scheme), so skews are reliable to about ±0.01 and the ratio to about ±0.05; skew and curvature are central differences at a half-width $h = 0.10$, and the ratio uses the ±5% strike pair, because Monte Carlo noise rules out Table 21.3's $h = 0.01$; it too is not the quotient of the skew columns.

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Table 21.4: One-month and one-year at-the-money level, skew, and curvature as each parameter is varied from the rough Bergomi base case ($H = 0.10$, $\eta = 1.9$, $\rho = -0.9$, flat $\xi_0 = 0.04$).

	One month			One year			
	ATM (%)	Skew	Curv.	ATM (%)	Skew	Curv.	**Ratio**
Base	17.8	−0.63	4.3	16.3	−0.31	0.0	2.6
$H = 0.25$	18.6	−0.73	0.1	15.1	−0.39	−0.3	2.0
$H = 0.45$	19.5	−0.47	−0.3	15.3	−0.36	−0.2	1.3
$\eta = 1.0$	19.3	−0.51	−0.1	18.6	−0.19	−0.1	2.7
$\eta = 3.0$	15.2	−0.47	13.3	12.8	−0.30	2.1	2.1
$\rho = -0.5$	18.3	−0.34	5.9	17.1	−0.17	0.9	2.4
$\rho = 0$	18.6	0.00	6.7	17.7	0.00	1.2	—

Two of the three read exactly as they did for Heston. *ρ sets the tilt*: at $\rho = 0$ the skew is zero at both maturities and the smile is symmetric. *η sets the curvature*, and drags the level with it as σ_v did: from $\eta = 1.0$ to 3.0 the one-month curvature goes from −0.1 to 13.3 and the at-the-money level from 19.3% to 15.2%.

The third has no counterpart in Heston. From $H = 0.10$ to 0.45 the one-month/one-year skew ratio falls from 2.6 to 1.3, where (21.8) predicts $12^{1/2-H}$: 2.70, 1.86 and 1.13, which is close, given that (21.8) describes the skew as maturity shrinks and is only approximate between a month and a year (Heston, a diffusive-variance model with $H = 1/2$, gives 1.77, not 1). The one-year skew, meanwhile, stays between −0.31 and −0.39 across the same three rows. H is a term-structure parameter: it decides how the skew is distributed across maturities, not how much of it there is at a year, and Heston has nothing that does that job. Two further points attach to the table. The parameters interact: the standard deviation of log variance over a horizon t is $\eta\, t^H$, so with the same η a rougher process swings more within a month, and the one-month level falls as H falls; over a year those swings largely average out, so the rough row's average variance is the least dispersed and its one-year level the highest. And on index surfaces three parameters, with the forward variance curve taken from the market, fit better than Heston's five (Bayer, Friz, and Gatheral, 2016), which is the strongest argument for the model.

Estimating H. The number did not come from option prices. Gatheral, Jaisson, and Rosenbaum built a daily series of realized volatility, σ_t on day t being the square root of that day's realized variance from five-minute returns, took its logarithm, and for each lag Δ from one to about thirty days computed the average p-th power of the change over that lag,

$$m(p, \Delta) = \frac{1}{n} \sum_t \left| \log \sigma_{t+\Delta} - \log \sigma_t \right|^p$$

over the n pairs of days in the sample, for powers p from about 0.5 to 3. Averaging over start days measures a single quantity because the processes they fitted, fractional Brownian motion and its slowly mean-reverting version, have stationary increments: a change over Δ days has the same distribution whichever day it starts on. Self-similarity then gives $m(p, \Delta) \propto \Delta^{pH}$, so a regression of $\log m$ on $\log \Delta$ has slope pH; the slopes came back proportional to p, as the model requires, with H of order 0.1, and nowhere near 0.5, for every index examined. Two cautions attach to the number. Realized variance is an estimate of the variance, not the variance itself, and measurement noise in it makes a series look rougher than it is; how much of the measured roughness is noise is still debated, as the end of this section explains. And the estimate is made under the physical measure; the H that prices options is read off the skew term structure under the risk-neutral measure, and the two roughly agreeing, as the start of this section noted, is the evidence, not an assumption.

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Where the frontier is. Two developments since 2019 matter. Rough Heston replaces the exponential decay of a variance shock in Heston (Figure 21.5) with the power-law kernel. Because the model is still affine (the logarithm of its characteristic function is linear in today's forward variance curve, as Heston's is in v_0 above), its characteristic function survives: it solves a fractional Riccati equation numerically rather than in closed form, but one integral still prices a vanilla and the machinery of Section 21.4 carries over. And the joint calibration problem, one model fitting the index smile and the VIX smile of Chapter 22 at the same time, long defeated the classical models and was narrowed but not closed by rough volatility. Among the models that fit both, path-dependent volatility makes today's volatility a weighted function of the index's own past path, its recent returns and their recent squares, rather than of a separate factor. That is Section 19.9's leverage effect, seen from the modeling side.

Two caveats follow. Rough models fit vanilla surfaces very well and their advantage for pricing exotics is real but smaller than the vanilla fit suggests. And the choice of H is not innocuous for anything sensitive to the very short end, where the model's predictions are most distinctive and market data are thinnest.

What is still contested. Each leg of the case for roughness has been questioned since 2022. On S&P 500, Euro Stoxx 50 and DAX surfaces, El Amrani and Guyon (2023) find that the power law fits from about a month to three years but not below a month, where a skew rising toward a finite limit fits better. Cont and Das (2024) show that realized volatility estimated from a process that is not rough can still measure as rough, so the time-series estimate of H may be partly an artifact of estimation error. And the quintic Ornstein–Uhlenbeck model (a polynomial in one mean-reverting Gaussian factor) of Abi Jaber, Illand, and Li (2023) fits SPX and VIX smiles jointly with a single Markov factor and no rough paths, its steep short-end skew coming from fast mean reversion and a large vol-of-vol. Rough volatility remains the most economical account of the skew term structure; whether volatility is in fact rough is open.

21.9 LOCAL-STOCHASTIC VOLATILITY

Desks divide the work. Local volatility serves payoffs the vanilla surface alone determines, and jump models serve mainly the short-dated skew; for path-dependent equity and foreign-exchange exotics, what desks run is neither the model of Section 21.2 nor that of Section 21.4. It is both:

$$dS = (r - q)S\,dt + L(S,t)\sqrt{v}\,S\,dW_1 \tag{21.10}$$

with v following a stochastic-volatility process as in (21.4). The function $L(S,t)$, the **leverage function**, is chosen so that the model reproduces the vanilla surface exactly.

The stochastic part supplies realistic dynamics; the leverage function absorbs whatever is left over so the fit is exact. What follows says what L is, what the one free choice in the construction does to a price, and who gets to make that choice.

What the leverage function is. A model reproduces the vanilla surface if and only if the average of its instantaneous variance, conditional on the spot at each date, equals the Dupire local variance at that spot, the result [TN21] derives from Gyöngy's theorem; local variance is therefore itself an average, of the instantaneous variance over the paths that pass through (S,t). For (21.10) the instantaneous variance is $L^2 v$, so

$$L^2(S,t) = \frac{\sigma^2_{\text{loc}}(S,t)}{\mathbb{E}\left[\, v_t \mid S_t = S \,\right]} \tag{21.11}$$

The numerator is the variance the surface demands at (S,t); the denominator is the variance the stochastic factor delivers there on average. The leverage function is the part of the smile the stochastic factor did not explain: above one where the stochastic model under-delivers, below one where it over-delivers, and exactly one where it already fits. The equation is circular,

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since the conditional expectation on the right depends on the paths and the paths depend on L. It is solved forward in time, one step at a time, by a forward partial differential equation or by a particle method that estimates the expectation from the simulated paths as they advance. [TN21] sets out the fixed point and both ways of solving it.

The mixing parameter. Scale the volatility of volatility, $\sigma_v \to \zeta\,\sigma_v$ with $\zeta \in [0, 1]$, and recalibrate L at each setting. At $\zeta = 0$ the variance is deterministic, the denominator of (21.11) is just $v(t)$, and $L = \sigma_{\text{loc}}/\sqrt{v}$: pure local volatility. At $\zeta = 1$ the stochastic factor explains as much of the smile as it can and L carries only the residual. Every setting reproduces the vanillas exactly, which is what makes ζ different from every other parameter in this chapter: it is not a *fit* parameter. It is a choice about dynamics that the vanilla market cannot make for you, and Figure 21.6 shows what it does to L. The vanilla surface there is the whole implied-volatility surface, every strike and maturity, that Heston produces at the base case of Figure 21.2, and the stochastic factor is that same base-case variance process, so at $\zeta = 1$ the stochastic model already fits and L equals one everywhere—the check that the construction is working. At $\zeta = 0$ the leverage function is the entire local-volatility shape, 1.51 at a spot of 75 falling to 0.57 at 115, and at $\zeta = 0.5$ it is flatter but not a midpoint: just below the money it dips below both, because the half-strength stochastic factor delivers more at-the-money variance than the surface wants.

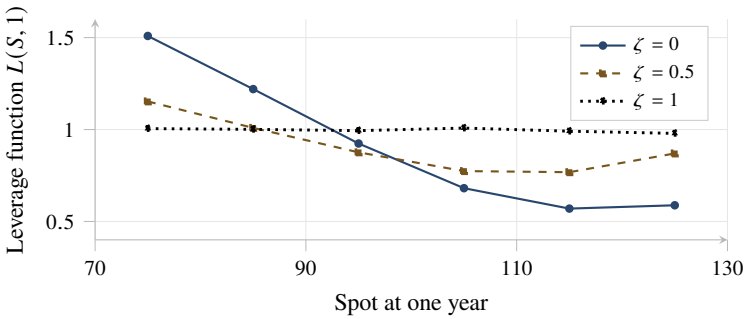


Figure 21.6: The leverage function at one year on Heston's own surface, at three settings of the mixing parameter.

Example 21.5 The mixing parameter as a dial

Heston's base case generates the vanilla surface; the local-stochastic model (21.10) is calibrated to it at five settings of ζ by the particle method, and prices a one-year up-and-out call, strike 100, barrier 110, monitored daily (extinguished when the spot first reaches 110); P(survive) is the share of paths that never touch the barrier. The particle method runs 80,000 Monte Carlo paths; the vanilla check is the one-year implied volatility the model returns at three strikes, against Heston's 23.37%, 16.94% and 13.33%.

	One-year vanilla fit				
Mixing ζ	$K = 80$	$K = 100$	$K = 120$	**Up-and-out 110**	**P(survive)**
0	23.35%	16.96%	13.30%	0.420	44.8%
0.25	23.34%	16.94%	13.29%	0.463	45.7%
0.5	23.27%	16.89%	13.28%	0.572	47.7%
0.75	23.21%	16.85%	13.23%	0.707	50.2%
1	23.24%	16.85%	13.23%	0.845	52.7%

The vanilla columns are flat. Every row matches the target to within 0.2 volatility points, which

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is the noise of the simulation and of averaging within spot ranges: the fit is exact at every setting, as the construction promises. *The barrier column doubles.* From $\zeta = 0$ to 1 the price of the same option, against the same vanilla surface, goes from 0.420 to 0.845, and it moves monotonically, so any value between the two can be produced by a setting of the dial. Both ends are legitimate calibrated models. Nothing the vanilla market quotes distinguishes them. The two ends are the two models of Example 21.7, whose independent simulation prices the same option at 0.4244 and 0.8402; the differences from 0.420 and 0.845 here are within the two runs' standard errors (about 0.005 and 0.008).

Section 21.11 returns to this option and explains the direction: a stochastic-volatility path that ends between the strike and the barrier is far more likely to have stayed below the barrier.

The mixing parameter is therefore not a nuisance. *It is the model risk, made into a dial*: a desk that prices a barrier at both ends of it has measured its own exposure to the modeling choice without building a second model, and the range (here a factor of two) is the number to take to a risk committee, not either end of it.

Who sets the dial, and how. The model is the standard on two kinds of desk, and they set ζ differently. In foreign exchange, where local-stochastic volatility has been the standard model for barrier and touch products since the mid-2000s, one-touch and double-no-touch options (which pay a fixed amount if the exchange rate touches a level, or if it never leaves a range, before expiration) trade actively enough to calibrate to: the desk picks the ζ that reproduces their prices, and the market has, in effect, priced the dial. In equities there is no such instrument. Barriers, autocallables and cliquets trade over the counter and are priced by the same desks that would need them as inputs, so ζ is set by judgment (from the historical volatility of implied volatility, from how the forward smile has behaved, or by house policy), and Section 21.11 will argue that the honest output is then the range rather than a point. The products most sensitive to the setting are the ones whose value depends on the forward smile: autocallables and the snowballs of Chapter 35, whose knock-in put has its barrier where the put is in the money, the reverse-barrier case on which Example 21.7 shows the models disagree.

Living with it. Two practical consequences follow from (21.11). Because L is a function of today's surface, it is not calibrated once and kept; it is re-derived whenever the surface moves, which in practice means daily, and the recalibration is the expensive part of running the model. And because the denominator is a conditional expectation estimated from finitely many paths, L is noisiest in the wings, where paths are few. A leverage function that looks jagged there is an estimator artifact, not a discovery, and practitioners smooth it before use. The delta and vega the model produces lie between local volatility's and Heston's, so the hedge, not only the price, moves with ζ.

21.10 CALIBRATION IN PRACTICE

Every model above has to be fitted before it can be used, and the fitting is where most of the practical damage is done. Three decisions matter more than the choice of model.

What to fit to. A surface contains hundreds of quotes of wildly different quality: the at-the-money one-month option trades continuously, the five-year 10-delta put trades twice a month. Fitting to all of them equally lets the least reliable quotes drive the parameters.

The standard remedies are to fit implied volatilities rather than prices (equivalently, to divide each price error by the option's vega) and to weight each quote by its liquidity, through bid–ask width or open interest. Dividing by vega is the most common choice, because it measures every error in volatility points, the unit in which options are quoted and bid–ask spreads are set, so a wing option's error counts as much as an at-the-money one. Weighting volatility errors *by* vega does the opposite, turning them back into price errors, which the at-the-money options

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dominate.

Fit quality against parameter stability. A model refitted every day to the whole surface will produce parameters that jump. Some of that is real; most of it is the optimizer trading one parameter against another along the flat directions of the objective function, which the overlapping effects in Table 21.2 create.

The consequence is not cosmetic. A book revalued with jumping parameters generates phantom profit and loss, and a desk cannot distinguish a genuine mark-to-market move from a calibration artifact. Standard practice is therefore to penalize movement away from the previous day's parameters (a regularization term), accepting a slightly worse fit in exchange for parameters that mean something over time.

Which risk the calibration is for. A model fitted to the whole surface is fitted to no part of it especially well. If the book's exposure is concentrated in one-year 25-delta puts, that is what the calibration should target, and the fit elsewhere matters little.

The objective should match the hedge: a model that reproduces every quote and mishedges the one position on the book has failed. Example 21.6 shows the objective alone moving the fit.

Example 21.6 Two fits, one surface

Fit Heston to the jump model's surface of Example 21.3, which no Heston parameters reproduce, at one month, three months and one year, strikes 80 to 120. One fit minimizes squared price errors; the other first divides each price error by the option's vega, which to first order makes it an implied-volatility error.

Price errors are dominated by at-the-money options, which have the largest vega, so a given volatility miss costs most there. The one-month 80 put is worth \$0.05, so the first fit leaves its implied volatility 7.08 points too low while missing the one-month at-the-money option by 0.12 points. The second cuts the wing miss to 4.01 points at a cost of 0.19 points at the money, and the average absolute one-month miss falls from 1.72 points to 1.11. Weighting volatility errors *by* vega instead nearly gives back the first fit (6.67 points in the wing).

Neither is right in general. The first is right for a desk whose book is at-the-money straddles; the second for a desk whose book is in the wings, like the structured-product issuers of Section 19.8. The objective encodes an assumption about the book; choose it rather than inherit a software default.

21.11 MODEL RISK MADE CONCRETE

Section 19.12 asserted that two models fitting today's surface perfectly can price the same barrier option far apart. That assertion is now checkable, because Sections 21.2 and 21.4 give two models that agree on every vanilla by construction.

Example 21.7 The same vanillas, two barrier prices

Take the Heston parameters of Figure 21.2, and derive the local-volatility surface from Heston's own vanilla prices by (21.2). The two models now agree on *every* European option: priced by simulation, the one-year 100-strike call comes to 6.766 under Heston and 6.767 under local volatility, against the semi-analytic 6.751, equal within Monte Carlo error.

Now price barrier options under each, one year, monitored daily, using the *same random numbers* for both models so that the difference is measured directly rather than as the difference of two noisy prices. This is Section 20.10's common-random-numbers technique.

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Barrier	Heston	Local vol	Difference (± s.e.)	Relative
Down-and-out call, strike 100				
80	6.5286	6.5821	+0.053 ± 0.015	+0.8%
85	6.2180	6.2962	+0.078 ± 0.015	+1.3%
90	5.5388	5.6138	+0.075 ± 0.015	+1.4%
Up-and-out call, strike 100				
130	5.8138	5.5787	−0.235 ± 0.015	−4.0%
120	4.0046	3.4193	−0.585 ± 0.013	−14.6%
115	2.4071	1.7138	−0.693 ± 0.010	−28.8%
110	0.8402	0.4244	−0.416 ± 0.005	**−49.5**%

On the down-and-out the two models agree to within one and a half percent, a difference a desk could ignore. On the up-and-out they disagree by nearly half the price.

So model risk is not a property of "barrier options." It is a property of where the payoff lives relative to where the models disagree. A down-and-out call struck at the money pays off in a region well above the barrier where the two models describe the distribution almost identically, because the vanillas pin it down. A reverse knock-out (the case Section 19.12 described, as this up-and-out call is) pays off only in the narrow band between the strike and the barrier, and that band is exactly where the surface constrains the models least: no vanilla option isolates the joint behavior of "finish in this band" and "never touch the top of it."

The direction is worth stating carefully, because a plausible-sounding argument gets it backward. One might reason that local volatility is low at high spot (Example 21.1 gives 12.0% at a spot of 110 a year out, against 16.2% at 100), so paths approach the upper barrier slowly, knock out less, and the option should be worth *more*. It is worth less, and the premise is false: local-volatility paths knock out more often (44.8% survive against 52.7% in Example 21.5). Both models put the same probability on every terminal level, so what matters is survival *given* where a path ends. Of the paths finishing between 108 and 110, 39% have never touched 110 under Heston and 14% under local volatility; of those finishing between 100 and 102, 77% against 57%. Under Heston some paths are quiet and some wild, and the quiet ones reach the band without touching the barrier; local volatility reproduces the same terminal distribution with one volatility at each spot and date, so it has no quiet paths to offer.

So the vanilla market does not price this product and no recalibration will make it. The number to report is the range the mixing parameter of Example 21.5 produces, and this table says before the trade which products have a wide one.

Case Study 21.2 When the model choice showed up in the profit and loss

Autocallables are the clearest case of this chapter's model choice paid for in cash. The structure, described in Section 12.12 and Chapter 35, embeds a down-and-in put that the investor sells to the issuer and a series of discrete observation dates. Its value depends on three things the vanilla surface does not determine: how the smile evolves between observation dates, volatility near the barrier, and, for notes on several underlyings, how their correlation rises in a decline.

Issuers hedge these books dynamically, and the hedge is computed from a model. Through the 2010s many issuers priced them on local volatility, because it fitted the vanillas exactly and the fit was auditable. Example 21.2 identifies the flaw in that choice. The issuer is long the knock-in put but short the coupon and early-redemption payments, which are digitals, and a steeper forward skew raises the value of both, so the sign of the net exposure depends on the barriers and the coupon. On many notes the digitals dominate, the issuer loses when forward skew steepens, and local volatility, which predicts

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a flattening forward smile, *understates* what the issuer owes; on any note it puts the vega profile in the wrong maturities.

The consequence stays hidden while the market is calm and appears when the index falls toward the knock-in region, when the forward smile matters most and the misspecified hedge has to be adjusted quickly. Section 18.12 described the resulting flow (issuers buying index futures into the decline, then selling at the barrier, after which their buying weakens); the model question is why the hedge needed adjusting then rather than earlier.

Desks moved to local-stochastic volatility for these books, and the mixing parameter of Section 21.9 is now a first-order risk number rather than a technical setting. They moved because the simpler model's error had a price, discovered in a drawdown.

The general lesson is that a model's failure mode determines what it may price: local volatility fits and mis-forecasts, stochastic volatility forecasts and mis-fits, and products that depend on both need the two combined, with the remaining uncertainty in the combination's free parameter.

21.12 CHOOSING A MODEL

The decision is not about realism. It is about which failure the trade is exposed to.

Marking a vanilla book → no model at all: interpolate the surface of Section 19.12.

Pricing a barrier or an autocallable → local-stochastic volatility, and report the range across the mixing parameter, since here the choice has a material cost.

Forward-starting products, cliquets → stochastic or rough volatility. Local volatility gets the forward smile structurally wrong (Example 21.2), and the forward smile is the product.

Very short-dated options → a smile fitted directly on a business-time clock (Section 23.9), with rough volatility or jumps as the check that some process could produce it. Classical stochastic volatility cannot make the skew, and Chapter 23's 0DTE market lives entirely in this region.

Swaptions and rate smiles → SABR, which Section 29.9 sets out and calibrates.

Two habits follow, parallel to Chapter 20's.

Calibrate to what you will hedge with. Fit to the instruments that will actually be in the hedge, and check the fit there rather than in aggregate.

Price it twice, with different models. The spread is not noise to be averaged away; it is the number the trade's model risk is measured in, and Example 21.7 shows how far two models can differ while agreeing on everything observable.

21.13 WHERE THIS CAME FROM

The Black–Scholes–Merton formula was accurate enough that its failures took more than a decade to become commercially serious, and the crash of 1987 made them so. The equity smile appeared and did not go away.

Three answers came within seven years, two of them already in print before the crash, and all three survive. Merton (1976) had already added jumps, which produce a smile at short maturities and were the natural response to a market that had just gapped. Hull and White (1987), the standard reference and still the paper to read for the structure of the problem, made volatility itself stochastic, as did Wiggins (1987) and Scott (1987) independently. Stein and Stein (1991) added a tractable mean-reverting form, and Heston (1993) displaced them all—not because its dynamics are better but because its characteristic function is known in closed form and the price is therefore a single integral. Carr and Madan (1999) made that computational accident decisive by showing how to invert such transforms fast.

The other answer was to give up choosing the dynamics. Dupire (1994), with Derman

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and Kani (1994) and Rubinstein (1994) arriving from trees, showed that exactly one local volatility surface reproduces every quoted vanilla. This chapter is largely about the price of that exactness. Bates (1996) combined jumps with stochastic volatility, the form in which jump models are now mostly used. Hagan and co-authors (2002) then gave SABR, whose asymptotic formula made a stochastic-volatility smile quotable from a formula and became the market's language for rates. And Gatheral, Jaisson, and Rosenbaum (2018) reopened the question by measuring that volatility is rougher than any of these assume.

Section 21.11's method, reading the spread between models fitted to one vanilla surface as the model risk, is Hull and Suo's (2002). They applied it to the implied volatility function model (local volatility) and found what Section 21.11 finds: agreement on every quoted vanilla leaves some exotics nearly fixed and barrier options far apart. The contribution was to make model risk a measurement rather than a warning, and it recurs in Chapters 34 and 39.

SUMMARY

A volatility model must fit today's surface and predict how it moves. Nothing in the vanilla market pins down the second requirement, and that degree of freedom is what every model in this chapter spends differently.

Local volatility makes σ a deterministic function of price and time, read directly off the surface by Dupire's formula, so the fit is exact by construction and the market stays complete. The local volatility is about twice as steep in strike as the implied volatility, because an implied volatility is an average of the local volatilities a path visits, and averaging flattens. Its failure is dynamic: the local-volatility skew decays steeply with maturity, so the model predicts a nearly flat forward smile where the market's is roughly stationary, and forward-starting products are mispriced and hedged with the wrong vega.

Heston makes variance a mean-reverting square-root diffusion correlated with the price. Correlation produces the skew, vol-of-vol produces the curvature, and mean reversion flattens both. No parameter moves one feature alone, so calibration is a genuine optimization and individual fitted parameters are not separately meaningful. Jumps produce short-dated skew no diffusion can, but that skew collapses with maturity faster than the market's does.

The at-the-money skew follows a power law $\psi(T) \sim T^{-0.4}$ from about a month to a few years; below a month the evidence is disputed. Heston saturates instead, rising too little toward the short end, and decays too fast at the long end: missing both ends at once is a structural failure, not a parameter problem. Bates, which runs a variance diffusion and jumps together, brings the short end close to the market's and pays for it at the long end. Eight parameters place the fit; they do not produce the law.

Rough volatility is the most economical resolution, though each leg of its case is now contested. Log-volatility behaves like a fractional Brownian motion with $H \approx 0.1$, which yields $\psi(T) \sim T^{H-1/2} = T^{-0.4}$ from a single parameter, and roughly the same H appears in realized volatility and in option prices. Rough Bergomi, given the market's forward variance curve, needs three parameters and fits index surfaces better than Heston. What made it usable was fast simulation, Markovian approximations that restore a state vector, and neural-network calibration: the Markov assumption it displaced was a computational convention, never an economic claim.

Exotics desks run local-stochastic volatility, a stochastic variance process multiplied by a leverage function calibrated to make the vanilla fit exact, and the mixing parameter between the two is the model risk turned into a dial. Two models agreeing on every vanilla can still price the same barrier far apart, because they differ in how the price travels to each terminal distribution, so the output is a range, not a price. Calibration is a modeling choice too: the objective decides which quotes the fit honors and should match the book's hedge, and parameters that stay stable

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from day to day are bought with a worse fit.

FURTHER READING

Gatheral's *The Volatility Surface* treats local and stochastic volatility together and is the natural next step from this chapter. Dupire (1994) is short and worth reading in the original for the argument rather than the formula. On Heston, the original 1993 paper sets out the model, but the numerically important material (branch cuts in the characteristic function, and the formulation that avoids the "little Heston trap") is in Albrecher and co-authors and in Gatheral, and any implementation that skips it can produce silently wrong prices at long maturities. Hagan and co-authors (2002) is the SABR paper; the wing corrections that repair its negative densities are later and scattered. For rough volatility, Gatheral, Jaisson, and Rosenbaum (2018) is the empirical paper that started it and is explicit about what was measured; Bayer, Friz, and Gatheral (2016) introduce rough Bergomi, and El Euch and Rosenbaum (2019) give rough Heston its characteristic function, which is the paper to read for how the affine machinery survives roughness. Bates (1996) is the paper that adds jumps to stochastic volatility, and Duffie, Pan and Singleton (2000) the general affine framework that explains why its characteristic function factorizes. For the local-stochastic model, Guyon and Henry-Labordère's *Nonlinear Option Pricing* (2013) is the account of the particle method by the people who made it practical, and is clear that the difficulty is the conditional expectation rather than the calibration. Guyon and Lekeufack (2023) is the path-dependent volatility paper behind the frontier remark in Section 21.8. On calibration itself, Christoffersen and Jacobs (2004) is the paper to read before choosing an objective function: fitting the same model to dollar price errors and to implied-volatility errors produces different parameters and different out-of-sample performance, which is why Section 21.10's weighting question is a modeling decision rather than a numerical one. The literature on Markovian approximations and on neural-network calibration is recent and still moving.

PRACTICE QUESTIONS

21.1 Explain why the denominator of Dupire's formula (21.2) is proportional to the implied density, and what that implies about when a local-volatility model exists.

21.2 Explain the "rule of two" using the averaging argument, and state whether a steep local-volatility surface is evidence of a calibration error.

21.3 Using Example 21.2, explain why a local-volatility model misprices a forward-starting option and in which direction.

21.4 A Heston fit returns $\rho = -0.05$. What shape of smile does this imply, and in which market would you expect to see it?

21.5 Explain why raising σ_v changes the at-the-money level as well as the curvature, and what this implies about interpreting a single fitted parameter.

21.6 State what jumps produce that a diffusion cannot, and state the corresponding defect in the term structure they generate.

21.7 Define the at-the-money skew $\psi(T)$, state the empirical power law it obeys, and explain why a classical stochastic-volatility model cannot reproduce a skew that keeps steepening as maturity shortens.

21.8 State the relationship between the Hurst parameter and the skew term structure, and give the value of H that reproduces the observed exponent.

21.9 Explain what the leverage function in a local-stochastic volatility model does, and what

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the mixing parameter measures.

21.10 Two models reproduce every quoted vanilla price and disagree on a barrier. Explain how this is possible without either committing an arbitrage, and state what a desk should report as a result.

21.11 Explain why a calibration objective is usually stated in implied volatility (price errors divided by vega) rather than in price, what a price-error objective would be blind to, and what a desk gains and gives up by penalizing day-to-day parameter movement.

FURTHER QUESTIONS

21.12 Example 21.1 finds a local-to-implied slope ratio of 1.75 rather than exactly 2. Derive the averaging argument that produces the factor of two, identify the approximation under which it is exact, and explain what features of the surface make the realized ratio depart from it.

21.13 Section 21.3 argues that local volatility gets the forward smile wrong while fitting today's surface exactly. Construct two processes that produce identical terminal distributions at every maturity but different forward smiles, and use them to explain precisely what information about the process the vanilla surface does not contain.

21.14 Table 21.2 shows that a rise in κ and a fall in σ_v both flatten the smile. Show that they do so through different mechanisms, explain why this makes them hard to identify jointly from one maturity, and describe what data would separate them.

21.15 Section 21.7 shows Heston too flat at the short end and decaying too fast at the long end simultaneously. Explain why adding parameters cannot fix both, and state what this implies about the practice of calibrating a single model to an entire surface.

21.16 Equation (21.8) gives $\psi(T) \sim T^{H-1/2}$. Explain why $H < 1/2$ makes the skew diverge as $T \to 0$, discuss whether an infinite short-maturity skew is economically sensible, and state what the model is really claiming about very short-dated options.

21.17 The case study in Section 21.8 argues that the Markov assumption in volatility modeling was a computational convention rather than an economic claim. Identify two other modeling conventions in this book that may be of the same kind, state what would have to become cheap for each to be abandoned, and explain how one would tell the difference between a convention and a claim.

21.18 Table 21.3 holds the variance rate fixed while the jump parameters change, and the one-year skew barely moves while the one-month skew moves a great deal. Using the fact that a jump component's contribution to skewness decays like $T^{-1/2}$ while a correlated diffusion's grows with maturity at first, explain why, and state what this implies about which options a Bates calibration should be run on to pin down the jump parameters.

21.19 Example 21.5 shows the up-and-out price rising monotonically with the mixing parameter while the vanilla fit is unchanged. Explain the mechanism, state why a foreign-exchange desk can calibrate the mixing parameter from market prices while an equity desk cannot, and describe what an equity desk should report instead of a single price.

21.20 Example 21.7 produces two barrier prices from models that agree on every vanilla. Explain why the local-volatility price is the lower of the two, construct a portfolio of vanillas that would partially hedge the difference, and explain why no portfolio of vanillas can hedge it completely.

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Chapter 22

Volatility Products: VIX, Variance Swaps, and Dispersion

Section 12.4 said that a trader who wants volatility exposure must delta-hedge an option and that the residual is a bet on realized against implied. Chapter 18 came within a step of saying what that residual is: a delta-hedged position is long gamma and short theta, and Section 18.4 found that it comes out level on a day the stock moves by one daily standard deviation at its implied volatility. Section 22.1 takes the step. Both descriptions make volatility something you get *left with* after removing what you did not want.

This chapter is about the instruments that deliver it directly. A variance swap pays realized variance against a fixed rate with no delta to hedge and no dependence on where the price moved. A VIX futures contract references an index computed from option prices. A dispersion trade isolates correlation. Each of these turns a by-product into a contract, and each is imperfect in a different and instructive way.

Why anyone wants such contracts shapes the whole market. Everyone who holds equities is short volatility whether they know it or not: drawdowns arrive when volatility rises, and the hedges that would soften them cost the most at exactly that moment. So there is a natural buyer of volatility, the holder of equity, and a natural seller, the intermediary or fund paid to bear what the holder wants to shed. Section 19.10 put the price of that transfer at two to four points of implied over realized volatility on equity indices and called it the volatility risk premium. The instruments here let that premium be bought and sold on its own, without the delta, the strike and the rebalancing that come with an option. That is their appeal. It is also why the sellers tend to be crowded, and why, with investors in aggregate structurally short volatility through their equity, the episodes in this chapter (2008, February 2018, March 2020) recur in the same shape. The route follows the order in which the market built the instruments: the variance swap and its caps (Sections 22.2 to 22.5), VIX, its futures and its products (Sections 22.6 to 22.8), then the dealer's view (Sections 22.9 to 22.13).

22.1 THREE WAYS TO OWN VOLATILITY

Delta-hedged options. Buy an option and hedge the delta. Section 18.1's dealer, hedging a written call, stood on the other side of this trade. The hedge cancels the delta term of Chapter 18's expansion, equation (18.1), and what is left are its gamma and theta terms: the buyer earns $\frac{1}{2}\Gamma(\Delta S)^2$ on each day's move and pays theta, whose cost over the day, with the interest term dropped, is $\frac{1}{2}\Gamma S^2\sigma^2\,\Delta t$ at the implied volatility σ. With the day's return written $\Delta S/S$, the day's profit is

$$\frac{1}{2}\Gamma S^2\Big[(\Delta S/S)^2-\sigma^2\,\Delta t\Big]$$

realized less implied variance, weighted by $\frac{1}{2}\Gamma S^2$. Summed over the position's life this is Section 12.4's residual, but weighted by gamma: it depends on *where* the underlying moved, not only how much, and two paths with the same realized volatility pay differently. The identity

holds for every option (a put, a butterfly, the straddle traders use because it starts close to delta-neutral); only Γ changes. It also holds σ fixed: if implied volatility moves, the vega term of equation (18.1), Section 18.5's vega profit or loss, is added.

Variance swaps. Contract directly on realized variance. There is no delta, no gamma weighting and no rebalancing. The exposure is exactly what the name says.

VIX futures. Trade an index constructed from option prices. This is the most accessible and the least direct: a VIX futures contract is not a claim on volatility but a claim on the *level of a computed index at a future date*, and Section 22.7 shows how far apart those can be.

The three are not substitutes, and most of the trouble in this market comes from treating them as though they were.

22.2 THE VARIANCE SWAP

A variance swap pays, at maturity,

$$N_{\text{var}}\left(\sigma_R^2 - K_{\text{var}}^2\right) \tag{22.1}$$

where σ_R^2 is the annualized realized variance of log returns over the life of the contract, K_{var} is the strike, quoted as a volatility, and N_{var} is the variance notional, paid per variance point (volatilities in percentage points, so a realized variance of 0.04 enters as $20^2 = 400$).

The vega notional convention. Variance notionals are awkward to think in, so the market quotes **vega notional**, the slope of the payoff in realized volatility, taken at the strike, so that a one-point move in realized volatility earns N_{vega} to first order. Since the slope of (22.1) at $\sigma_R = K_{\text{var}}$ is $2K_{\text{var}}N_{\text{var}}$,

$$N_{\text{var}} = \frac{N_{\text{vega}}}{2K_{\text{var}}} \tag{22.2}$$

so that, by (22.2), the payoff (22.1) becomes $N_{\text{vega}}(\sigma_R^2 - K_{\text{var}}^2)/(2K_{\text{var}})$. A desk saying it is "short \$100,000 of vega at 20" means it loses roughly \$100,000 if realized volatility comes in at 21 rather than 20—*roughly*, because the payoff is not linear in volatility: the exact loss is $100{,}000 \times (21^2 - 20^2)/(2 \times 20) = \$102{,}500$, and the extra 2.5% is the convexity that is the subject of Section 22.4.

Realized variance is defined, not observed. The contract specifies the observation times (daily closes), the return convention (log returns), the annualization factor (252), and, importantly, whether the mean is subtracted:

$$\sigma_R^2 = \frac{252}{n}\sum_{i=1}^{n}\left(\ln\frac{S_i}{S_{i-1}}\right)^2 \tag{22.3}$$

with S_0 the close on the trade date and S_i the official close on the ith of the n scheduled observation dates. That sum is the contract. It is what the confirmation says and what settlement is computed from, and no integral appears in any document. No mean is subtracted in it, and that is standard practice: it is what lets the option strip of Section 22.3 replicate the payoff, and it makes the figure differ slightly from a sample variance. These are not details: two contracts with the same strike and different terms are different contracts, and the cap that Section 22.5 describes, standard on single-stock contracts and found on index contracts since 2008, is one such term.

22.3 REPLICATING VARIANCE

The result that makes this market work is that variance can be replicated from vanilla options held *statically*: no model, no volatility forecast, and nothing traded as the price moves except a stock position of constant dollar size. The route is Chapter 12's. Section 12.8 built any

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piecewise-linear payoff from a bond, a forward, and options at its kinks; a smooth payoff is the limit of piecewise-linear ones, so it is built the same way from a continuous strip of options. Apply that to the smooth payoff $-2\ln(S_T/F)$, the **log contract**, with T the contract's maturity in years and F today's forward price for that date. Two facts combine, and each takes a paragraph.

Delta-hedging a log contract earns exactly realized variance. Let the price follow

$$\frac{dS}{S} = \mu_t\,dt + \sigma_t\,dW$$

with μ_t any drift and σ_t any volatility process (constant, stochastic, path-dependent); nothing below depends on which. Itô's lemma applied to $\ln S$ gives

$$d(\ln S) = \Big(\mu_t - \frac{1}{2}\sigma_t^2\Big)dt + \sigma_t\,dW, \qquad \text{so} \qquad \frac{dS}{S} - d(\ln S) = \frac{1}{2}\sigma_t^2\,dt$$

The gap between the simple return and the log return accumulates half the variance, whatever the drift and whatever the path. Integrating over the life of the contract and multiplying by $2/T$,

$$\frac{1}{T}\int_0^T \sigma_t^2\,dt = \frac{2}{T}\left[\int_0^T \frac{dS}{S} - \ln\frac{S_T}{S_0}\right] \tag{22.4}$$

The left side is realized variance, in continuous time: the quantity the contract's sum (22.3) approximates with daily closes, and the limit that sum tends to as the observations become dense, by the $dW^2 = dt$ of Section 14.7. The right side is a trading strategy: the integral is the profit, before financing and dividends, of holding $1/S_t$ of the stock at every instant (a continuously rebalanced position with a constant dollar exposure), and the logarithm is the payoff of a contract on $\ln S_T$. Realized variance is therefore two things a desk can hold, and no probability has entered.

The gamma of the log contract does the same job one rebalance at a time. Its payoff is $-2\ln(S_T/F)$, and its value at time t, V_t, is that payoff's risk-neutral expectation discounted to t. Take σ constant for the moment and write $F_t = S_t e^{(r-q)(T-t)}$ for the forward at t, with r the risk-free rate and q the dividend yield. Under the risk-neutral measure $\mu_t = r - q$, and integrating the $d(\ln S)$ equation above from t to T gives $\ln S_T$ the mean $\ln F_t - \sigma^2(T-t)/2$, so

$$V_t = e^{-r(T-t)}\,E_t^{Q}\left[-2\ln\frac{S_T}{F}\right] = e^{-r(T-t)}\left[-2\ln\frac{F_t}{F} + \sigma^2(T-t)\right]$$

and its gamma is

$$\Gamma = \frac{\partial^2 V_t}{\partial S^2} = \frac{2e^{-r(T-t)}}{S^2}$$

which is the payoff's own second derivative, $2/S^2$, multiplied by the discount factor. The dividend yield sits inside F_t and does not reach the second derivative. The daily gamma gain of Section 18.3 is therefore

$$\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}\cdot\frac{2e^{-r(T-t)}}{S^2}\,(\Delta S)^2 = e^{-r(T-t)}\Big(\frac{\Delta S}{S}\Big)^2$$

the present value of the day's squared return, whatever the level of S. That is precisely the weighting that makes the summed gamma gains independent of where the moves happened: an ordinary option's gamma is concentrated near its strike, and the same move earns it more there than elsewhere. Carried to expiration at the risk-free rate, each day's gain is worth exactly the squared return, and that is the e^{rT} in (22.5) below: the strip's option prices are present values, and the replication multiplies them back up.

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The log contract's payoff is replicated by a strip of options weighted $1/K^2$. So the contract can be built from what trades. Section 12.8 built piecewise-linear payoffs by placing an option at each kink, in a quantity equal to the change in slope there. A smooth payoff $f(S_T)$ is the limit: a kink at every strike, with slope change $f''(K)\,dK$. Expanding around the forward, with puts below it and calls above so that every option used is out of the money,

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)\,(K - S_T)^+\,dK + \int_F^\infty f''(K)\,(S_T - K)^+\,dK$$

For $f(S_T) = -2\ln(S_T/F)$ the pieces are

$$f(F) = 0, \qquad f'(F) = -\frac{2}{F}, \qquad f''(K) = \frac{2}{K^2}$$

a short position in $2/F$ forward contracts, which costs nothing to enter, and $2\,dK/K^2$ of an option at every strike.

Write σ^2_{var} for the fair variance swap rate: the strike K^2_{var} at which (22.1) is worth zero today, which is the risk-neutral expectation of realized variance, $\sigma^2_{\text{var}} = E^Q[\sigma^2_R]$. Take that expectation of (22.4). Under the risk-neutral measure the stock's expected return is $r - q$, so the stock position's expected profit is

$$E^Q\left[\int_0^T \frac{dS}{S}\right] = (r - q)\,T = \ln\frac{F}{S_0}$$

and combining it with the log term gives

$$\sigma^2_{\text{var}}\,T = -2\,E^Q\left[\ln\frac{S_T}{F}\right]$$

so the fair total variance $\sigma^2_{\text{var}}T$ is the expected payoff of the log contract. Take that expectation term by term in the expansion above: the forward term has zero expectation, and each option's expected payoff is its price grown at the risk-free rate, $e^{rT}P(K)$ or $e^{rT}C(K)$, with $P(K)$ and $C(K)$ today's prices of the put and call struck at K and expiring at T. The result is

$$\sigma^2_{\text{var}} = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right] \tag{22.5}$$

which is (19.2), now derived rather than asserted. Figure 22.1 shows what the $1/K^2$ weighting looks like across the strip: the low strikes carry most of the weight (a strike at 50 four times the at-the-money strike's, a strike at 200 a quarter of it), which is why a variance swap is far more sensitive to the downside wing than a trader used to at-the-money vega would expect.

Example 22.1 Building the strip, and what truncation costs

Take a flat 20% surface, one year, $S_0 = 100$, $r = 2\%$, no dividends (forward 102.02); the answer must be exactly 20%.

Build the strip from options out of the money relative to the forward, at strikes every 5 points from 50 to 200, weight each by $\Delta K/K^2$, and apply (22.5):

$$\text{replicated volatility} = \mathbf{19.96\%}$$

against a true 20.00%. The obvious suspect for the 4 basis points of error is the truncation of the integral at 50 and 200, and the table shows it is the wrong one.

The table widens the range and, in its last row, narrows the spacing:

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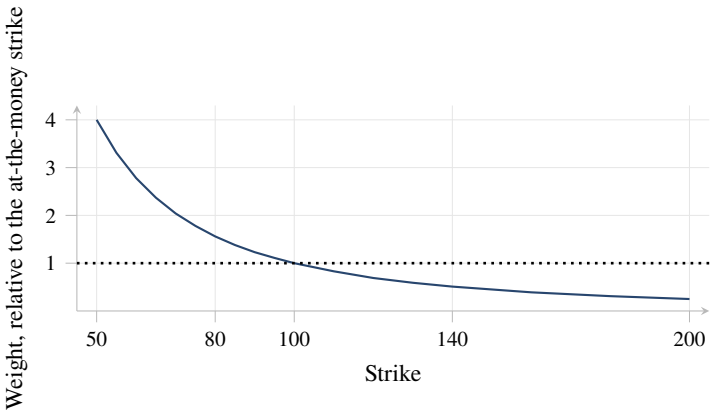


Figure 22.1: Weight on each option in the variance strip, relative to the at-the-money strike ($S_0 = 100$).

Strike range and spacing	Replicated volatility
50–200, step 5	19.956%
30–300, step 5	19.956%
20–500, step 5	19.956%
10–800, step 2	20.015%

Two lessons follow. Widening the range beyond 50–200 buys nothing, because at a 20% volatility a one-year put struck at 50 is worth a fraction of a cent and contributes nothing despite its large weight. But the spacing matters, and not because the integrand is hard. The error depends on where the grid meets the forward, 102.02, at which the strip switches from puts to calls and the out-of-the-money price peaks in a kink: a strike sitting on that peak credits its whole cell with the peak value, so a strike exactly on the forward overshoots by +10 basis points, while this 5-point grid, straddling it, falls 4 short. The error shrinks with the square of the spacing, so the 2-point grid, whose strike at 102 is close to that grid's worst placement, is within 1.5 basis points.

The ranking of the two errors reverses when volatility is high. At 60% volatility the options beyond 50 and 200 are worth something real, the truncation error dominates, and Section 19.3's warning applies with force: the strip stops where the quoting rule stops, not where the integral does.

22.4 WHY VARIANCE AND NOT VOLATILITY

The obvious contract would pay realized *volatility*, which is what people think in. The market trades variance instead, and the reason is Section 22.3: variance is replicable and volatility is not. Volatility swaps exist, and trade mostly in foreign exchange, where they are often quoted in place of variance swaps, but they are priced off variance swaps with an adjustment, and Example 22.2 shows that the adjustment is large enough to matter.

Example 22.2 The variance–volatility spread

Suppose realized volatility over the period is, under the risk-neutral measure, lognormal with mean 20%. The variance swap strike is $\sqrt{E^Q[\sigma_R^2]}$ and the volatility swap strike is $E^Q[\sigma_R]$, and by Jensen's inequality the first exceeds the second.

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Vol-of-vol ν	Variance swap	Volatility swap	Spread
20%	20.40%	20.00%	0.40 pts
30%	20.92%	20.00%	0.92 pts
50%	22.66%	20.00%	2.66 pts
70%	25.55%	20.00%	5.55 pts

Since for a lognormal $E^Q[\sigma_R^2] = (E^Q[\sigma_R])^2 e^{\nu^2}$, the spread is $\bar{\sigma}\left(e^{\nu^2/2} - 1\right)$, where $\bar{\sigma}$ is the mean of realized volatility over the period and ν the standard deviation of its logarithm over the same period (a vol-of-vol measured over one period), so the spread grows quickly with ν. If the period's realized volatility has a log standard deviation of 40% to 60%, as is realistic for a one-year index swap, the variance swap strike sits one and a half to four volatility points above the volatility swap strike on the same underlying.

Two consequences follow. A variance swap strike is not a volatility forecast. Quoting a 22.66% variance swap and calling it "the market's expected volatility" overstates the forecast by 2.66 points. This is not a further overstatement on top of the excess over at-the-money volatility that Section 19.3 attributed to the skew: at-the-money implied volatility sits close to the volatility swap strike (a little below it when spot and volatility move inversely), so that excess is largely this same gap.

And the spread is itself a traded view. Buying a volatility swap against a variance swap is a position in vol-of-vol, one of the few ways to trade it without options on variance.

22.5 THE CONVEXITY IS NOT FREE

The same convexity that makes the strike high makes the payoff asymmetric, and Figure 22.2 shows it directly: the variance swap's payoff curves away from the straight line a volatility swap would pay, touching it at the strike and lying above it everywhere else. The gap grows with the square of the distance from the strike, equally on both sides, but realized volatility can rise much further above the strike than it can fall below it.

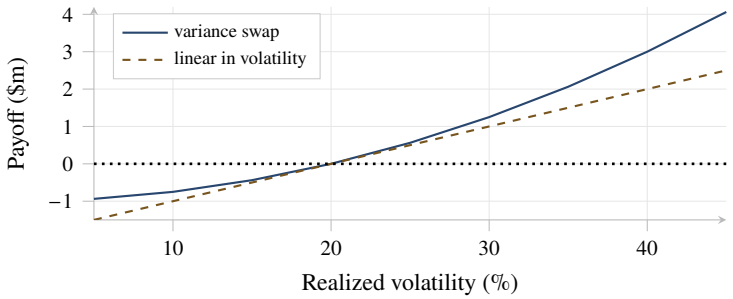


Figure 22.2: Payoff of a long variance swap struck at 20%, vega notional $100,000 per point.

Example 22.3 The buyer's convexity, and the seller's problem

A long variance swap struck at 20% with $100,000 of vega notional pays:

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Realized	Payoff	If linear in vol	Convexity gain
10%	−$750,000	−$1,000,000	+$250,000
15%	−$437,500	−$500,000	+$62,500
20%	$0	$0	$0
25%	+$562,500	+$500,000	+$62,500
30%	+$1,250,000	+$1,000,000	+$250,000
40%	+$3,000,000	+$2,000,000	+$1,000,000

The buyer gains from convexity in both directions, which is why the strike is above the volatility swap's. The seller pays for it in one direction only in practice, because realized volatility cannot fall below zero but can rise without limit: at 40% realized the seller loses $3 million on a position whose best realistic gain is the $750,000 at 10%. The seller's loss has no upper bound, while its gain can never exceed $1 million, the payoff at zero realized volatility.

In 2008 realized volatility on equity indices reached levels several times the strikes at which variance had been sold, and sellers, many of them dealers who treated the short variance as an offset to the options they were long, took losses far beyond anything the vega notional suggested.

The market's response was structural. A **cap** on realized volatility, conventionally 2.5 times the strike, had been standard on single-stock variance swaps from the start, because their gaps make the tail far worse; after 2008 that market nearly stopped trading even with its caps, and capped index contracts appeared too, though uncapped ones still trade. A capped variance swap is a variance swap less a call on realized variance struck at the cap, so it is no longer statically replicable by (22.5); the cap must be priced with a model of how forward variance moves, such as the rough-volatility models of Section 21.8.

So the instrument that was attractive because it needed no model acquired one, as a direct result of the payoff its replication made possible.

Case Study 22.1 March 2020, and what a cap is worth

The 2008 lesson of Example 22.3 was tested twelve years later, and where the repair had been written into the contract, it held. Realized volatility on major equity indices ran above 80% annualized over the worst month of the pandemic sell-off, against variance swap strikes of 12% to 18% set in the weeks before it. On uncapped terms (the pre-2008 terms, on which much of the index market still traded), that is a loss no seller's capital was sized for.

Take a one-month swap struck at 15% just before the sell-off began, with $100,000 of vega notional, and suppose realized volatility over its life came in at 80%.

Uncapped, the seller loses

$$100{,}000 \times \frac{80^2 - 15^2}{2 \times 15} = 100{,}000 \times 205.83 = \$20.6 \text{ million}$$

on a position whose stated vega notional was one hundred thousand dollars. That factor of 206 is the whole problem with the instrument, and it involves no model error—it is what the payoff formula does when realized volatility comes in at more than five times the strike.

Capped at 2.5 times the strike, the loss stops at

$$100{,}000 \times \frac{37.5^2 - 15^2}{2 \times 15} = 100{,}000 \times 39.375 = \$3.9 \text{ million}$$

The cap reduced the loss by a factor of **5.2**, and, more to the point, it converted an unbounded exposure into a known one that could be capitalized against in advance. Two observations follow.

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The repair was structural, not behavioral. No one forecast the pandemic. What the market did after 2008 was change the contract so that a failure to forecast would not be fatal, which is a different and more reliable kind of prudence.

The cost was paid in model risk. Example 22.3 noted that a capped variance swap is no longer statically replicable. The sellers of capped variance who came through March 2020 with bounded losses did so holding an instrument they had to price with Chapter 21's machinery rather than with a strip of options. They traded a known unbounded risk for a smaller risk they could only estimate, and on this occasion that was clearly the right trade.

22.6 VIX: AN INDEX, NOT A PRICE

VIX is the variance swap rate for a 30-day maturity on the S&P 500, expressed as a volatility. But the published number is not (22.5). It is a discrete sum over a strip that Cboe, the exchange that publishes the index, selects by a rule fixed in advance, and every peculiarity of the index is a property of that rule rather than of the theory: why it sits above at-the-money implied volatility, why it depends on how thickly the wings are quoted, why it can move on a day when no volatility changed. So the rule is worth writing out.

The forward comes from the options, not from the index. For each expiration, with T its time to expiration and r the matching risk-free rate, let K^* be the strike at which the call and the put are closest in price. The construction sets

$$F = K^* + e^{rT}\big(C(K^*) - P(K^*)\big) \tag{22.6}$$

which is put–call parity read backward. Parity holds at every strike, so any listed strike would return the same F from error-free quotes; Cboe uses K^* because there both options are near the money and liquid, and the quote error in $C - P$ is smallest. K^* is a choice of estimator, not a requirement of the identity. Neither the S&P 500's level nor a dividend forecast enters. Whatever the market believes about carry over the next month is already inside the option prices, and (22.6) extracts it rather than assuming it, so the forward around which VIX is built is not the S&P 500's quoted level.

The strip is the out-of-the-money options around K_0. Let K_0 be the highest listed strike at or below F. The strip takes puts at strikes below K_0, calls above, and at K_0 itself the average of the two, so that every option used is out of the money except the pair at K_0, whose call is slightly in the money because $K_0 \le F$, and none is used twice. Each carries ΔK_i, half the distance between its neighbors. Walking outward from K_0 in each direction, the calculation stops after two consecutive strikes with no bid. Writing $Q(K_i)$ for the bid–ask midpoint of the option used at K_i (Cboe's notation, not the measure), the expiration's variance swap rate σ^2 (σ^2_{var} of Section 22.3) is

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{rT} Q(K_i) \;-\; \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 \tag{22.7}$$

The sum is (22.5) discretized. The subtracted term is not, and it is the one piece of the method that does not follow from Section 22.3.

Where that last term comes from. The replication splits the strip at F; the listed strip splits at K_0, which is a listed strike and so in general is not F. Expanding the logarithm about K_0 instead and taking risk-neutral expectations returns (22.5) with K_0 in place of F, plus the cost of having moved the split. With $x = F/K_0 - 1$, that cost is $2[\ln(1+x) - x] = -x^2 + O(x^3)$, the second-order remainder of a logarithm, and divided by T it is the subtracted term. Its size is bounded by $(\Delta K/K_0)^2/T$, a variance of about 0.00002 on the S&P's five-point grid near 4500, or about **0.005 VIX points** at a VIX of about 17.6, 24 days from expiration; Cboe carries

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it anyway because nothing guarantees a fine grid, and on a 25-point grid the bound is about 0.13 of a point. [TN22] derives it in full, states the exact correction the quadratic replaces, and measures it against the two larger errors that separate the published index from the variance swap rate.

Thirty days is a blend of two expirations. In general no option expires in exactly 30 days, so (22.7) is evaluated on the two nearest expirations that bracket it (the near term with more than 23 days to run, the next term with fewer than 37), and the results are interpolated *in total variance*, not in volatility:

$$\text{VIX} = 100\sqrt{\left[T_1\sigma_1^2\,\frac{N_2 - N_{30}}{N_2 - N_1} + T_2\sigma_2^2\,\frac{N_{30} - N_1}{N_2 - N_1}\right]\frac{N_{365}}{N_{30}}} \tag{22.8}$$

where σ_1^2 and σ_2^2 are (22.7) for the near and next expirations, N_1 and N_2 are the minutes from now to the near and next expirations, $N_{30} = 30 \times 1{,}440 = 43{,}200$ is the minutes in thirty days, $N_{365} = 525{,}600$ the minutes in a year, and $T_i = N_i/N_{365}$. The bracket is a linear interpolation of total variance to thirty days (variances over successive periods add; volatilities do not); the factor N_{365}/N_{30} annualizes it. The calculation counts minutes, not days, so the weights slide continuously through the session rather than stepping once a day. Weekly expirations have been eligible since 2014, which keeps the two brackets tight around 30 days; before that the window was wider and the interpolation did more work.

Example 22.4 Computing the index from a strip

Let the S&P 500 stand at 4500, with $r = 4.2\%$, options quoted on a five-point grid, and a downward-sloping skew: 17% at the forward, steepening to the downside. The near term expires in 24 days, the next in 31. The prices were generated with a 1.4% dividend yield, which the construction never sees.

Start with the forward. Equation (22.6) returns $F =$ **4508.29** for the near term, 8.3 points above the S&P's 4500 (the 24-day carry), so $K_0 = 4505$.

Then build the strip. The no-bid rule keeps 295 strikes, running from 3475 to 4945:

K	Option	ΔK	Price	$\Delta K e^{rT} Q/K^2$ ($\times 10^6$)
3500	put	5	0.060	0.024
4000	put	5	2.302	0.721
4300	put	5	20.540	5.570
4505	put/call avg.	5	78.347	19.355
4700	call	5	11.785	2.675
4900	call	5	0.207	0.043
Total over all 295 strikes				1020.478

Note how far the strip reaches on each side: 22.9% below the forward but only 9.7% above. The skew keeps deep puts bid long after the corresponding calls have lost their bids, so the strip is asymmetric before any truncation rule is applied.

Assembling (22.7):

$$\frac{2}{T}\sum \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) = 0.0310395, \qquad \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 = 0.0000081$$

so $\sigma_1^2 = 0.0310314$ and $\sigma_1 =$ **17.62%**. The correction moved the answer by 0.0023 VIX points, consistent with the bound above: the term is a safeguard rather than a working part.

The 31-day expiration gives $\sigma_2 =$ **17.82%**. The weights in (22.8) are ratios of the N, so their unit cancels; taken at a moment when the expirations are exactly 24 and 31 days away, $N_1 = 24 \times 1{,}440$ and $N_2 = 31 \times 1{,}440$ minutes, the weights are $(31 - 30)/(31 - 24) = 1/7$ and

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6/7, and

$$\text{VIX} = \mathbf{17.80}$$

closer to the 31-day figure than to the 24-day one, because 30 days is closer to 31.

Now thin the wings. Raise the price below which an option shows no bid, so the near-term strip, for example, runs from 3890 to 4835 instead of 3475 to 4945 (190 strikes rather than 295). VIX falls to **17.70**. No option changed its implied volatility; the index fell a tenth of a point because fewer options were quoted.

Three consequences follow from that experiment and from (22.7) and (22.8). The $1/K^2$ weighting makes VIX exceed the at-the-money implied volatility, because low strikes carry more weight and the skew makes them expensive. Apply (22.5) to the one-year skew of Figure 19.1 (standing in for a 30-day skew), interpolating between its plotted strikes and holding the smile flat beyond them, with zero rates, over strikes from 50 to 200: it returns 24.38% against an at-the-money implied volatility of 20.00%, while a flat 20% smile returns exactly 20.00%. The 4.38-point gap is the skew premium, and it widens as the skew steepens. On the strikes that skew actually quotes, 70 to 130, the answer falls to 21.99% and the flat control to 19.56%, so truncation costs something even with no skew at all.

The truncation rule makes VIX depend on quoting depth. The calculation stops including strikes when it meets two consecutive strikes with no bid, so on a quiet day the wings are thin, the sum is truncated early, and the index is measured on a narrower strip than on a busy one. That is worth a tenth of a point in Example 22.4, and more when volatility is high and the wings matter most. On August 5, 2024, before the US open, VIX printed 65.73 while S&P 500 futures were down about 3 to 4%: the formula was reading the midpoints of very wide pre-open SPX quotes. It closed at 38.57. That is Example 22.4's dependence on quotes rather than trades, at full scale and in the other direction: wide quotes inflated the midpoints the sum reads.

And the 30-day constant maturity is an interpolation. Two listed expirations bracketing 30 days are blended in total variance, so VIX refers to a synthetic maturity that no traded option has. As one expiration rolls out of the window and another rolls in, the weights in (22.8) shift, and around the roll the index can move for reasons that have nothing to do with volatility.

Together these make VIX a well-defined number that is not the price of anything that can be held. The strip that defines it today is not the strip that defines it tomorrow, so no position kept over time tracks it and it cannot be arbitraged against.

22.7 VIX FUTURES AND THE TERM STRUCTURE

A VIX futures contract is marked to market daily, like any futures contract, against a settlement price taken from trading at the close. Its final settlement, at expiration, is in cash against the **Special Opening Quotation** (SOQ): the VIX formula applied to the opening prices, set in a special opening auction on the settlement morning, of SPX options that expire exactly 30 days later, so no interpolation is needed; settlement falls on the Wednesday 30 days before that standard (third-Friday) SPX expiration. Like the index options of Section 10.3, the contract settles to a value that is computed rather than traded, and it can differ noticeably from both the previous close and the index a few minutes later. So the SOQ fixes the futures price on the final day, and the price converges to it. Before then the price is *not* tied to the index's current level, and the reason is worth being precise about.

A VIX futures contract is not a forward on VIX. Chapter 5's cost-of-carry argument requires that the underlying can be bought and stored, and VIX cannot be held. So there is no carry arbitrage linking the futures price to the spot index. A futures contract costs nothing to enter and is marked to market daily, so under the risk-neutral measure its price has zero expected growth (Section 13.10) and converges to the index at expiration: the futures price is the risk-neutral

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expectation of VIX_T, the index's value at the futures expiration T. Since VIX_T^2 is a 30-day variance rate and the square root is concave, Section 22.4's Jensen effect places the futures price below the square root of the forward variance swap rate, the fair variance rate for the 30 days that follow T (Section 22.12).

The practical consequence is the term structure. Because volatility mean-reverts (Section 19.6), the curve is usually upward sloping when spot volatility is low (which is most of the time) and inverts in stress: in Chapter 5's terms, contango in calm markets and backwardation in stressed ones. Mean reversion also makes the futures move less than the index: between the two curves of Figure 22.3 the index rises from 15.5 to 40 but the front contract only from 16.4 to 33, about two-thirds as far, and the five-month contract under a quarter as far. A hedge sized on spot VIX therefore under-hedges, and more so the further out the contract. Besides the issuers of Section 22.8's products, who hedge in the first two contracts, the curve's main traders are hedge funds positioning on its slope.

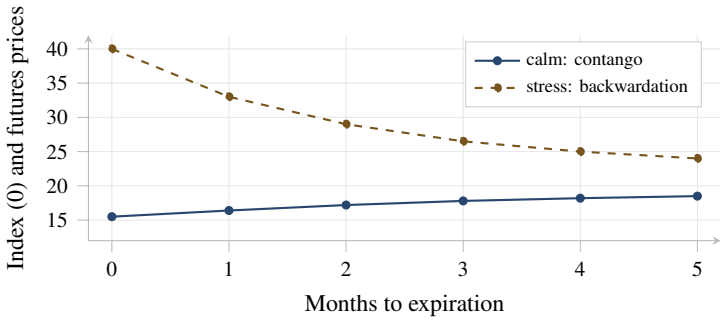


Figure 22.3: Representative VIX futures curves. Illustrative.

Example 22.5 What contango costs a long position

Using the calm curve of Figure 22.3: the front month is 16.4 and the second month 17.2.

A fund holding constant one-month exposure must roll: it buys the second month at 17.2, and one month later, if the curve has not moved, that contract has become the front month and is marked at 16.4.

$$\text{loss} = 17.2 - 16.4 = 0.8 \text{ points}, \qquad 0.8/17.2 = \mathbf{4.65\% \text{ per month}}$$

Compounded, $(1 - 0.0465)^{12} - 1 = -43.5\%$ **per year**, with the VIX index unchanged throughout.

This is Section 5.11's roll yield with no carry behind it, large because the VIX curve is steeper than most futures curves. It has two consequences, one for each side of the trade.

Long VIX futures products are structurally poor hedges to hold. An investor who buys volatility exposure as portfolio insurance and holds it for a year pays something like 40% of the position for the privilege, and the payoff must beat that before the hedge has done anything.

And short VIX futures products earn that roll as a carry, which is the volatility risk premium of Section 19.10 collected through a different instrument. Section 22.8 is about what happened to the people collecting it.

The premium and the roll are the same thing seen from two sides. Mean reversion alone would tilt the curve upward from a low spot level, but a futures price that merely forecast the index would cost a long holder nothing on average; the curve slopes up by *more* than that because sellers demand compensation for bearing the risk, and that excess is both the roll cost to the buyer and the compensation to the seller. Neither is a market inefficiency, and a strategy built on "harvesting contango" is not exploiting anything—it is being paid to take a risk that occasionally arrives.

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22.8 VIX EXCHANGE-TRADED PRODUCTS

Section 18.12 counted February 2018 among the episodes in which hedging moved the market. Here is the mechanism from the product's side.

An **inverse VIX product** is an exchange-traded note (a bank's debt) or fund whose value moves opposite, day by day, to an index of a rolling position in the first two VIX futures contracts: it rises 1% for every 1% that index falls, and falls 1% for every 1% it rises. It is in effect a short position in those contracts, packaged so that a retail investor can collect the roll carry of Example 22.5 without a futures account, and by early 2018 the two largest, XIV and SVXY, held about $3 billion between them. Offering $-1\times$ exposure *daily* means the product must rebalance every day to restore that exposure, exactly as the leveraged funds of Section 14.9 do.

Example 22.6 Why an inverse product must buy into a rally

A fund with net asset value (NAV) A maintaining $-1\times$ exposure is short futures with notional A. Suppose futures return R over the day.

The NAV becomes $A(1-R)$, so the new target short is $A(1-R)$.

The existing short has grown to $A(1+R)$ of exposure.

So the fund must buy

$$A(1+R) - A(1-R) = \mathbf{2AR}$$

of futures at the close. It buys as the price rises. The rebalance is destabilizing, and it is Section 18.12's short-gamma feedback in a different form.

With roughly $3 billion in inverse VIX products, the rebalancing works out as follows:

Futures move	NAV after	Buying required
+10%	$2.70bn	$0.60bn
+25%	$2.25bn	$1.50bn
+50%	$1.50bn	$3.00bn
+96%	$0.12bn	$5.76bn

The demand grows as the fund shrinks. At a 96% move the products have lost 96% of their capital and must buy $5.76 billion of futures (nearly twice their original size) into a market that is already moving against them.

The buying was due in the fifteen minutes before the futures close, when the market normally trades a small part of its daily volume. The rebalance was not a large order in a deep market; it was a large order in a thin one, announced in advance.

Figure 22.4 puts the two quantities on one axis, and the crossing is the whole problem. The purchase the fund owes the market rises with the move while the fund itself shrinks, and on any day the futures rise by more than a third the buy exceeds everything the fund has left. February 2018 lay beyond the point where those lines cross.

Case Study 22.2 February 5, 2018, from the volatility products

The S&P 500 fell about 4% that day—a bad day, not a historic one. VIX futures roughly doubled.

For a $-1\times$ product, a 100% move in the underlying futures is total loss: the NAV multiplier is $(1-R)$, and at $R = 1$ it is zero. The largest inverse product, XIV, had a term in its prospectus allowing early termination if the value the issuer published during the day (its intraday indicative value) fell more than 80% from the prior close. It did. The note was redeemed early, and holders received $5.99 a note, against more than $100 a few days earlier. SVXY, a fund holding the futures directly rather than a bank's note and

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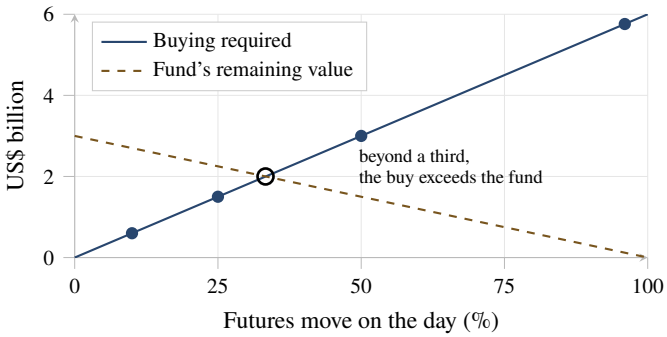


Figure 22.4: Daily rebalancing of a $-1\times$ inverse VIX product with \$3 billion of assets. Marks are the moves of Example 22.6.

carrying no such clause, lost about 90% on the same day but survived, and its sponsor cut the exposure to $-0.5\times$ three weeks later, where it remains. Three things about the episode deserve to be separated.

The product did what it said. A $-1\times$ daily inverse product loses everything on a 100% daily move, and the prospectus said so. This was not mis-selling in the narrow sense.

The rebalance amplified the move that destroyed it. Example 22.6 gives the size. The products had to buy into the close precisely because the price had risen, and their buying was a material fraction of the volume available. A product whose hedging requirement grows as it loses money is a product that helps arrange its own destruction.

The wider market was already fragile. Two years of low realized volatility had built large short-volatility positions well beyond the exchange-traded products: in the option-selling strategies of Section 19.10, in dealers' books, and in funds that size equity by its volatility (risk-parity and volatility-target funds), whose allocations fall mechanically when realized volatility rises. All of these were positioned the same way.

The general lesson is the one Section 18.12 drew from five separate episodes and Section 35.7 meets again in the autocallable market. Crowding plus mechanical hedging is the mechanism, and the instrument is incidental. 1987's portfolio insurance, 2018's inverse exchange-traded products, and 2024's snowball knock-ins are the same structure with different wrappers, and the recurring feature is that the hedging demand is a known function of the price, in 2018 a published one, calculable in advance by anyone who cared to.

22.9 THE PREMIUM, SEEN THROUGH EACH INSTRUMENT

Section 19.10 reported that implied volatility exceeds subsequently realized volatility by two to four points on equity indices. In variance-swap form that premium is measured as VIX^2 against the next 30 days' realized variance, as Carr and Wu (2009) measured it, and it varies over time. The three instruments of Section 22.1 all collect that premium, and they do not collect the same amount. The difference is a useful check on whether one understands what each contract is, and Example 22.7 applies it.

Example 22.7 One premium, three numbers

Suppose one-year implied volatility is 18% and realized comes in at 15%.

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A short delta-hedged straddle earns, by Section 22.1, the difference between implied and realized *variance*, gamma-weighted along the path:

$$18^2 - 15^2 = 324 - 225 = 99 \text{ variance points (squared volatility points)}$$

but the realization depends on where the underlying spent its time. If the stock spends the year far from the strike, the position has little gamma when the moves occur and collects a fraction of the 99.

A short variance swap at 18% earns exactly 99 variance points. Per unit of vega notional, by (22.2) the short position's payoff is $(K_{\text{var}}^2 - \sigma_R^2)/(2K_{\text{var}})$, here

$$\frac{18^2 - 15^2}{2 \times 18} = \textbf{2.75 volatility points}$$

with no dependence on where the moves occurred. It is less than the 3-point gap in volatility because the payoff is convex in σ_R (Section 22.4): the short gains less when realized volatility falls short of the strike than it would lose if it overshot by the same amount. The swap collects the straddle's quantity cleanly, which is the reason the contract exists.

A short VIX futures position earns something else. It collects the roll of Example 22.5 (0.8 points a month on the illustrative curve), which is a claim about the *shape of the curve*, not about the gap between one-year implied and one-year realized volatility. The two are related, but they are different contracts on different quantities and can move apart.

Instrument	What is collected	What can go wrong
Delta-hedged straddle	Implied − realized variance, gamma-weighted	Path: the gamma may not be where the moves are
Variance swap	Implied − realized variance, exactly	Convexity: the loss is quadratic
Short VIX futures	The roll down the curve	The curve inverts, and does so before realized volatility rises

The third row contains the trap. The VIX curve inverts on the *expectation* of volatility, which is to say immediately, while realized volatility rises only as the moves actually occur. A short VIX futures position therefore takes its loss at once, while a short variance swap's settlement loss accrues only as the moves occur; its mark-to-market also loses as implied variance rises, but the payoff waits for the realized path. Anyone treating the two as interchangeable will find the correlation between them is high in calm markets and much lower in exactly the week it matters.

22.10 WHAT GOES WRONG IN A VARIANCE SWAP BOOK

The replication of Section 22.3 is exact under assumptions, and a dealer's business is the gap between those assumptions and the world. Four gaps matter.

Jumps. The log-contract argument assumes a continuous path. A gap of log size J adds J^2 to realized variance (before annualization), while the replicating portfolio collects $2(e^J - 1 - J)$, which differs by about $J^3/3$ and falls short on a downward gap. The replication under-hedges precisely on the days that matter, and this is the residual that Section 21.5 said cannot be hedged by any position taken beforehand.

Dividends. Realized variance is computed from price returns, and a price drops on the ex-dividend date. Unless the contract adjusts for it (index contracts usually do not; single-stock contracts often add back the dividend actually paid), a large special dividend registers as volatility that never happened.

Single stocks. Section 22.5 recorded that single-stock variance swaps, capped from the outset, nearly stopped trading after 2008. The reason is the convexity of Example 22.3 applied to an underlying that can gap 40% on an earnings miss or a takeover bid: the seller's loss, even capped, is quadratic up to the cap in a move whose distribution has a much fatter tail than an

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index's. Dispersion desks that would naturally use single-stock variance often use straddles or other options on the constituents instead, because the capped swaps trade thinly and the cap cuts off the payoff on the large single-stock moves the component leg is long. That substitution reintroduces exactly the path dependence the instrument was designed to remove.

The strip is not the contract. The dealer holds a fixed portfolio of options; the client holds a claim on realized variance. If the underlying moves far enough that the strip's strikes are all deep in or out of the money, the replication's gamma no longer matches the log contract's, and the hedge degrades exactly when realized variance, and so the payoff owed, is largest. This is the same truncation issue as Example 22.1, now as a running risk rather than a pricing error.

None of these is a flaw in the theory. Each is a place where a contract defined to be model-free requires a judgment, and a variance-swap desk is in the business of pricing those judgments.

22.11 DISPERSION

Section 16.2 named dispersion traders and Section 19.7 gave the arithmetic: index and single-stock implied volatilities together imply a correlation. The dispersion trade of Example 19.4, selling index volatility against single-stock volatility, is a position in that correlation, and with variance swaps it can be constructed cleanly.

Take the numbers of Example 19.4, its implied volatilities taken as one-year variance swap strikes: 50 equally weighted stocks at 30%, the index at 18%, implying a correlation of 34.7%; the 2.6-point gap Example 19.4 found at a realized 25% is 2.38 points as a variance payoff (first row below). The weight on the component leg follows from Example 19.4's equal-weight form of (19.3). A parallel move in component variance σ_c^2 moves index variance σ_I^2, at fixed correlation, by the factor $\sigma_I^2/\sigma_c^2 = (0.18/0.30)^2 = 0.36$, so 0.36 units of variance notional spread equally over the 50 component swaps against 1 unit of index variance makes the position neutral to volatility and exposed only to correlation. In general the weight is $(K_I/K_c)^2$, the square of the index strike K_I over the component strike K_c.

Figure 22.5 plots the resulting profit against realized correlation, and the line is the point of the structure: it crosses zero at the implied 34.7% whatever the component volatility, and is linear in correlation with a slope set by component variance. Example 22.8 reads four outcomes off it and adds a calm one.

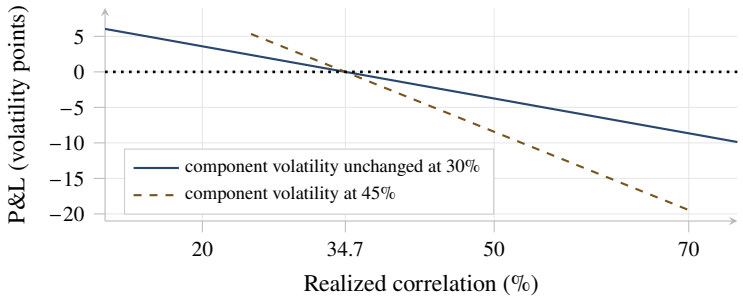


Figure 22.5: Profit on a variance-neutral dispersion trade against realized correlation.

Example 22.8 What dispersion earns and what it loses

The position is short 1 unit of index variance at 18% and long 0.36 units of component variance at 30%.

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Component vol	Realized corr	P&L	
30%	25.0%	+2.38	correlation below implied
30%	34.7%	0.00	correlation realizes at implied
30%	45.0%	−2.53	correlation rises, volatility unchanged
20%	20.0%	+1.60	calm: everything falls
45%	70.0%	**−19.46**	crash: volatility and correlation both jump

(P&L in volatility points per unit of index vega notional.)

The second row is the check. When realized correlation comes in exactly at the implied level, the trade makes precisely zero, whatever volatility does, as long as the components realize a common volatility. No parameter was tuned to produce it; it follows from the weighting, and it confirms that the position is a pure correlation bet.

The last row is the trade. The first row, correlation ten points below implied, earns 2.38 points; the crash loses 19.46. The ratio is one to eight, and the loss arrives in the same weeks as everything else in the portfolio.

Two causes combine. The correlation loss is scaled by realized component variance, and in a crash both rise together: at an unchanged 30% volatility, 70% correlation costs 8.65 points; at the 45% that accompanies it, 19.46. And correlation starting at 34.7% has nearly twice as far to rise as to fall, so the largest possible moves are the ones that run against the short index leg.

This is why the trade is described as "earning steadily and losing in blocks," and why it is run by desks that can size it against a capital base rather than by funds that must report monthly. Cboe publishes implied correlation as an index (COR1M, COR3M). In July 2024, with index returns concentrated in a few very large stocks, the three-month index fell to a record low and the trade had become crowded; when volatility spiked on August 5 (Section 22.6), correlation rose with it, in the direction of the last row.

Correlation swaps. A correlation swap pays realized average pairwise correlation against a strike directly, which sounds cleaner and is used less. The reason is replication: dispersion is built from variance swaps that can themselves be replicated from options, whereas a correlation swap cannot be replicated, and the dealer who writes one hedges it only approximately, with a dispersion position whose payoff is a variance-weighted proxy for the same correlation. The market trades the replicable proxy rather than the clean instrument, which is the same preference that made variance rather than volatility the standard contract. The buyers are mostly structured-product dealers, short correlation from worst-of notes (Section 34.11); the writers are hedge funds and dispersion desks already holding that view through variance.

22.12 FORWARD VARIANCE AND OPTIONS ON VOLATILITY

Section 21.8 observed that rough-volatility models specify the forward variance curve, today's fair variance rate for each future window, directly. That curve is a traded object, and this is where it trades.

A **forward-starting variance swap** contracts on variance over a future window, and it is replicated by a spread of two spot-starting variance swaps: long the longer, short the shorter, weighted by their maturities. Since both legs are replicable from options, so is the forward: the whole forward variance curve is spanned by today's vanilla surface, and it needs no storable underlying, which Chapter 5's forward prices do. Dealers use the forward swap to hedge the forward-volatility exposure of the cliquets and autocallables they issue (Section 34.12, Chapter 35), and volatility funds to take term-structure positions. Example 22.9 works one through.

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Example 22.9 Reading the forward variance off two spot swaps

Six-month variance swaps are quoted at 18% and one-year at 20%. Total variance to each date is TK^2_{var}: $0.5 \times 0.0324 = 0.0162$ and $1 \times 0.0400 = 0.0400$. The variance over the second half-year is the difference, spread over the half-year it covers:

$$\sigma^2_{\text{fwd}} = \frac{1 \times 0.0400 - 0.5 \times 0.0324}{1 - 0.5} = \frac{0.0238}{0.5} = 0.0476, \qquad \sigma_{\text{fwd}} = \mathbf{21.82\%}$$

The forward sits above both spot strikes, as a forward rate sits above both spot rates on an upward-sloping curve in Chapter 4, and for the same reason: a two-point rise in the term structure over a horizon that is only half the maturity has to be made up entirely in the later half. The check is the telescoping identity: the two half-year variances average to the one-year, $(0.0324 + 0.0476)/2 = 0.0400$.

Long one unit of the one-year swap and short half a unit of the six-month swap, in variance notional, is long 0.5 units of forward variance; scaled to one unit, long 2 against short 1, which is what "weighted by their maturities" meant.

What is *not* spanned is anything nonlinear in future variance. VIX futures (a square root of variance), options on realized variance and on VIX, and the caps of Section 22.5 all require a model of how variance moves, not merely its expectation. This is the boundary Chapter 21 drew.

VIX options are the largest such instrument. They are European and settle in cash to the same SOQ as the futures, and their forward is the futures price for the same expiration, not spot VIX, so put–call parity runs against the futures. Pricing them requires a joint model of the S&P 500 and of the volatility surface from which VIX is computed. Their implied volatilities carry an upward skew, the reverse of the equity skew, because VIX spikes up and not down, and the market pays more for the right tail of a quantity that spikes. Tail hedgers buy the out-of-the-money calls as crash protection; volatility sellers and dealers write them. The VIX formula applied to a strip of VIX options gives VVIX, the implied volatility of VIX itself. Fitting the SPX smile and the VIX smile with one model is the joint calibration problem of Section 21.8, and the steep VIX skew is what makes it hard: it demands a volatility of volatility the SPX smile alone does not reveal.

22.13 WHO TRADES WHAT

Buyers of volatility are the protection buyers of Section 16.2 using a different instrument. A pension fund buying VIX calls or entering a long variance swap is buying convexity against its equity book. It pays the premium of Section 19.10 and the roll of Example 22.5, and it does so knowingly: the alternative is holding less equity.

Sellers of volatility are collecting that premium. They divide into those who can hold the position through a drawdown (dealers with capital, insurers, and endowments) and those who cannot, which is most of the rest. Allianz Global Investors' Structured Alpha funds show a seller less hedged than it said: they sold index options against crash puts their investors were told they held, but the managers had bought cheaper puts further out of the money, and in March 2020 funds that had run more than $11 billion lost over $7 billion. The firm's US arm pleaded guilty to securities fraud in May 2022 and paid about $6 billion.

Dealers intermediate and end up long or short depending on what clients trade, hedging with the vanilla strip of Section 22.3; their business is the *difference* between that strip and the contract (Section 22.10).

Dispersion desks trade correlation, as Section 22.11 described, and are the natural counterparty to index protection buyers: the index put demand that steepens the index skew is what makes implied correlation high enough to be worth selling. Worst-of note issuance is a

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second source of that flow: the investors are long correlation, so the issuing dealers are short it (Section 34.11) and buy it back by buying index volatility against single-stock volatility, which lifts implied correlation further.

Volatility-target and risk-parity funds do not trade volatility products at all, but they are among the largest volatility *exposures* in the market, because their equity allocation is a decreasing function of realized volatility. A fund targeting 10% volatility holds equity in proportion $10\%/\sigma_R$, so realized volatility doubling from 12% to 24% halves its equity weight from 83% to 42%; a hypothetical $100 billion of such funds then sells about $42 billion of equity. They are mechanically short volatility without holding a single option, and Section 22.8's snapshot showed what that means when realized volatility jumps.

Investors, taken together with their equity, are structurally short volatility, and the premium is the price of shedding that risk. What the episodes in this chapter show is that the capacity to bear it is smaller than the appetite to collect it, and the gap is discovered at inconvenient moments.

22.14 WHERE THIS CAME FROM

The instruments in this chapter rest on a result that needs no model at all. Breeden and Litzenberger (1978) showed that the option chain reveals the terminal distribution outright, and Carr and Madan later showed that any payoff written on that distribution can be replicated from options. Neuberger's log contract (1994), the payoff whose delta-hedged profit is exactly realized variance, is what variance swaps were built on, and Demeterfi, Derman, Kamal and Zou (1999) wrote the replication out in the form desks use, the source of Section 22.3's $1/K^2$ weighting. The CBOE rebuilt VIX on the same replication in 2003, which is why the index is a variance calculation rather than an average of implied volatilities, and Carr and Wu (2009) then measured the premium the trade earns, Section 19.10's premium in variance terms.

SUMMARY

Volatility can be owned three ways (delta-hedged options, variance swaps, VIX futures), and they are not substitutes. The first pays realized minus implied variance weighted by gamma, so it depends on where the underlying moved; the second pays realized variance exactly; the third is a claim on the level of a computed index.

A variance swap pays $N_{\text{vega}}(\sigma_R^2 - K_{\text{var}}^2)/(2K_{\text{var}})$ on a realized variance the contract defines. It can be replicated statically from out-of-the-money options weighted $1/K^2$, because delta-hedging a log contract earns realized variance on every path and the log contract's payoff is replicated by that strip. Variance rather than volatility is traded because variance is replicable; the variance swap strike exceeds the volatility swap strike by an amount that grows quickly with the volatility of volatility, so it is not a volatility forecast. The same convexity makes the seller's loss quadratic: after 2008 the single-stock market, capped from the start, shrank, and capped index contracts appeared too; a capped contract needs a model, and March 2020 showed what a cap is worth.

VIX is the 30-day variance swap rate on the S&P 500 as Cboe's rule computes it: above at-the-money implied volatility because the $1/K^2$ weights load on the low strikes the skew makes expensive, dependent on quoting depth because of the truncation rule, and an interpolation between two expirations. It is a well-defined number that is not the price of anything that can be held. VIX futures are not forwards, because VIX cannot be held; rolling a long position on a contango curve loses steadily, at a rate that compounds over a year to a large share of the position, and that cost to the buyer is the premium to the seller.

An inverse VIX product with net assets A must buy $2AR$ of futures when futures return R, so its hedging demand grows as it loses money; on February 5, 2018 that demand overwhelmed

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the liquidity at the close. Dispersion sells index variance against component variance weighted $(K_I/K_c)^2$, makes zero when correlation realizes at the implied level, and loses many times its normal gain in a crash, because correlation rises when everything falls and the loss on it is scaled by component variance, which rises with it. The forward variance curve is spanned by today's vanilla surface; anything nonlinear in variance is not, and needs the models of Chapter 21.

FURTHER READING

Demeterfi, Derman, Kamal and Zou's paper on variance swaps is the original practitioner derivation of (22.5); it works the replication through in a way this chapter compresses. Carr and Lee survey what can and cannot be replicated in volatility, and their survey is the place to go for the volatility-swap adjustment stated properly rather than approximated. Cboe's VIX methodology documents the construction including the truncation rule and the roll, and is worth reading precisely because the index's peculiarities are all in the mechanics rather than the theory. On February 2018, the Bank for International Settlements' analysis in its March 2018 Quarterly Review is the standard account; the mechanism of Example 22.6 is arithmetic, and the question it addresses is how much of the day's move the rebalance actually caused. Bergomi's book (2016) is the practitioner reference for forward-variance models and for options on variance and on VIX.

PRACTICE QUESTIONS

22.1 A variance swap is struck at 20% with \$100,000 of vega notional. Compute the variance notional and the payoff at 35% realized, compare it with what a linear-in-volatility contract would give, and explain what determines the gap between a variance and a volatility swap strike.

22.2 Explain why delta-hedging a log contract earns realized variance regardless of the path, what feature of the log contract's gamma is responsible, and why the replication strip weights options by $1/K^2$ rather than equally.

22.3 Using Example 22.1, state whether widening the strike range or narrowing the strike spacing is the more effective way to improve a replication at low volatility, and explain why the answer changes at high volatility.

22.4 Six-month variance swaps are quoted at 25% and one-year variance swaps at 22%. Compute the forward volatility for the second half-year, explain why it lies below both spot strikes, and state the positions in the two swaps, in variance notional, that make up one unit of forward variance.

22.5 Explain why variance swaps, single-stock ones above all, are now often written with caps, and what the cap does to the replication argument.

22.6 Explain why VIX exceeds the at-the-money implied volatility of 30-day S&P options, and why a VIX futures contract is not a forward contract on VIX in the sense of Chapter 5.

22.7 Using the curve of Figure 22.3, compute the monthly and annualized roll cost of a constant one-month long VIX futures position, and state who receives it.

22.8 A $-1\times$ inverse VIX product has \$2bn of NAV. Compute the rebalancing trade required if futures rise 30% in a day, state its direction, and explain why the requirement grows as the product loses money.

22.9 Explain why a dispersion trade weighted $(K_I/K_c)^2$ makes exactly zero when realized correlation equals implied correlation, whatever the components' common volatility, and

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why it loses so much more in a crash than it earns in a normal quarter.

22.10 Implied volatility is 22% and realized comes in at 18%. Compute what a short variance swap earns in volatility points, and explain why a short delta-hedged straddle need not earn the same and why it typically earns less.

22.11 A special dividend of 8% is paid on a stock during the life of a variance swap that does not adjust for dividends. Explain what happens to measured realized variance and who gains.

22.12 Explain why dispersion desks often use straddles rather than variance swaps on single stocks, and what they give up by doing so.

FURTHER QUESTIONS

22.13 Example 22.1 finds that widening the strike range beyond 50–200 buys nothing at 20% volatility, while changing the spacing moves the result. Derive the truncation error and the discretization error, show how each scales with volatility and maturity, and determine the volatility at which they are equal for a one-year contract on strikes from 50 to 200 spaced 5 apart, with the forward on a listed strike.

22.14 Section 22.5 argues that capping a variance swap destroys its static replication and therefore requires a model. Decompose a capped variance swap into a variance swap and a call on variance, explain what information about the process the call requires that the swap does not, and connect this to Section 21.11's measurement of model risk.

22.15 Example 22.5 finds a 43.5% annual roll cost and asserts that this is compensation rather than an inefficiency. Construct the argument that a systematic short VIX futures position is a risk premium, then construct the argument that it is a crowded trade earning a rent, and state what evidence would distinguish them.

22.16 Example 22.6 shows that an inverse product's rebalancing demand is $2AR$. Derive the corresponding expression for a $+2\times$ leveraged long product, explain why both signs of leverage are destabilizing, and discuss whether a product rebalancing at a random time of day rather than at the close would be safer or merely less predictable.

22.17 Section 22.13 claims that volatility-target funds are mechanically short volatility without holding options. Formalize this: write the fund's equity allocation as a function of realized volatility, derive its sensitivity to a change in realized volatility, and express that sensitivity in units comparable to an option's vega. Then discuss what this implies about measuring the market's aggregate short-volatility position.

22.18 Example 22.8 shows a one-to-eight ratio between the first row's gain and the crash loss. Derive the general condition on the implied correlation under which a variance-neutral dispersion trade has positive expected value, given a joint distribution for realized correlation and component volatility, and explain why a desk might rationally run the trade even when that condition fails.

22.19 Example 22.9 reads a forward variance off two spot variance swaps. Repeat it for a curve that is *inverted* (six-month at 30%, one-year at 24%), and show that the forward strike falls below both spot strikes. Then find the six-month strike at which the implied forward variance would be zero, and explain what a quote above that level would mean and how it would be arbitraged.

22.20 Section 22.12 states that the whole forward variance curve is spanned by today's vanilla surface with no storable underlying required. Explain why the variance case needs no storage, identify the assumption that makes it work, and state what fails if that assumption is relaxed.

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Chapter 23
Short-Dated Options, Dealer Positioning, and Market Feedback

Every chapter so far has treated the option market as a place where risk is transferred. This one is about what happens when the transfer becomes large enough, and fast enough, that the hedging it requires becomes a force in the underlying market.

The occasion is a market structure that did not exist when the previous generation of textbooks was written. Options expiring the same day they are traded, the 0DTE options of Section 10.12, went from a curiosity to the majority of S&P 500 option volume in under a decade. Section 10.12 recorded the growth and deferred the mechanism here; Section 18.12 supplied the machinery, a dealer whose hedging moves the market it is hedging in; and Section 21.12 observed that this market lives where classical stochastic volatility cannot make a skew.

Three demands meet in the contract: a hedger whose risk lasts one session (a data release, a policy decision) and who used to buy a month of protection for it, a retail buyer wanting a leveraged intraday bet, and a systematic seller collecting the volatility risk premium of Section 19.10 at the shortest horizon there is (Section 23.13). Whether the hedging of their positions stabilizes or destabilizes the index depends on how the three net.

The chapter has an unusual shape, because the answer to its central question is that we do not yet know. It first calculates the mechanism by which 0DTE could destabilize the market (Sections 23.2 to 23.11), then weighs the evidence, which so far says mostly no, and why that is not reassuring (Section 23.12), and ends with what would have to go wrong (Section 23.14).

23.1 WHAT 0DTE IS, AND HOW IT GREW

An option expiring on the day it trades is not new—monthly and quarterly expirations have always had a final day. What is new is that there is one *every* day.

S&P 500 index options (SPX: European, settled in cash, $100 per index point, the weekly and daily series against the closing level) acquired weekly expirations progressively: Fridays first, then Wednesdays and Mondays in 2016, Tuesdays and Thursdays in 2022. Since then every trading day has been somebody's expiration, and a trader can put on a same-day position at any time.

The share of volume moved with it. On Cboe's figures, 0DTE contracts were a single-digit percentage of SPX option volume in the mid-2010s, passed half at the end of 2024, and ran close to three-fifths through 2025. The most heavily traded S&P 500 option is now one that will not exist tomorrow. Two things about that sentence are worth separating.

Volume is not risk transfer. A 0DTE contract has a fraction of the vega of a monthly option and lives for hours rather than weeks, so measuring the market in contracts overstates how much exposure has changed hands. That overstatement is the most common error in commentary on the subject.

But volume is hedging. Every contract traded is hedged by somebody, and the hedging of a 0DTE contract is concentrated into hours rather than spread over months. The flow is

compressed even where the risk is not, and compression is what makes a market impact.

23.2 WHY SHORT-DATED OPTIONS BEHAVE DIFFERENTLY

Everything peculiar about this market follows from what happens to the Greeks as $T \to 0$. Take the S&P 500 at 5000 with a 15% implied volatility (a normal quiet market), with a 4% rate and a 1.5% dividend yield. Time is counted in trading hours throughout, 6.5 to the day and 252 days to the year, because that is the clock a 0DTE option lives on; every number in the chapter is priced this way, and "30 days" means thirty trading days, about six calendar weeks.

Table 23.1: At-the-money call as expiration approaches. The last two columns are the delta-hedged profit per unit of the index on a 1% rise.

Time left	Price	Delta	Gamma	Gamma term	Exact P&L
30 days	110.47	0.5323	0.001534	1.92	1.89
7 days	51.58	0.5158	0.003188	3.98	3.90
1 day	19.10	0.5060	0.008443	10.55	9.60
4 hours	14.94	0.5047	0.010763	13.45	11.65
1 hour	7.43	0.5024	0.021528	26.91	17.55
15 minutes	3.71	0.5012	0.043056	53.82	21.25

Gamma at one hour is fourteen times gamma at thirty days (Table 23.1), and it keeps rising: at-the-money gamma diverges like $1/\sqrt{T}$, which Section 18.3's formula makes obvious once one looks at it. That one number gives the scale of the problem.

But the more important fact is not how large the gamma is—it is how *narrow* it is (Figure 23.1). The table's last two columns show it. Both are the hedged profit on a 1% rise, once as the gamma term $\frac{1}{2}\Gamma\,(0.01\,S)^2$ of the Taylor expansion and once by repricing the option; at one hour a 1% move is 2.7 hourly standard deviations, and gamma falls away as the index leaves the strike, to 3% of its at-the-money value by the end of the move. The exact profit is earned on a gamma that falls as the index moves, while the gamma term keeps the starting value, so the profit comes out a third below the gamma term, and at fifteen minutes it is two-fifths of it.

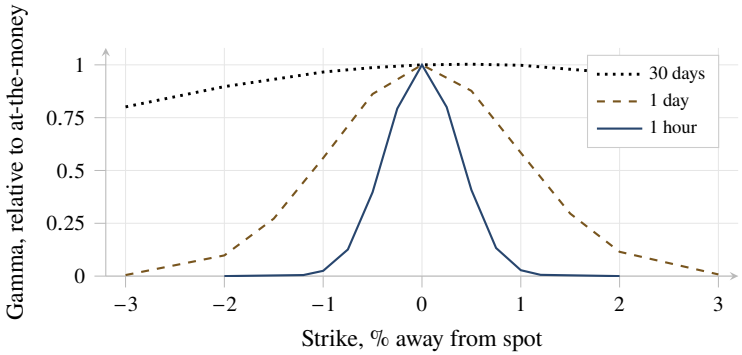


Figure 23.1: Gamma by strike, normalized to the at-the-money value.

Call the range of strikes over which gamma exceeds half its at-the-money value the **gamma band**. Its edges, which are symmetric in the logarithm of the strike about a point slightly above spot rather than about spot itself, close in as follows:

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Time to expiration	Edges of the gamma band, above / below spot
30 days	+6.76%/−5.52%
1 day	+1.13%/−1.09%
1 hour	+0.44%/−0.43%

This is the structural fact of the chapter. A month-dated book has gamma spread across a wide range of index levels, so a move of one percent changes the hedging requirement gradually. In its last hour a 0DTE book has four-fifths of its gamma within half a percent of spot and nearly all of it within one percent, so the hedging requirement is a step function of where the index happens to be. Move the index one percent and a dealer's entire gamma exposure can switch off, or switch on, or change sign.

23.3 THE BUYER'S ARITHMETIC

Before the market-structure question comes the individual trade. A 0DTE option is almost pure time value, and it has hours to live. Figure 23.2 traces that life through one session: with the index unchanged, the option loses value steadily for most of the day and then falls away in the final hour.

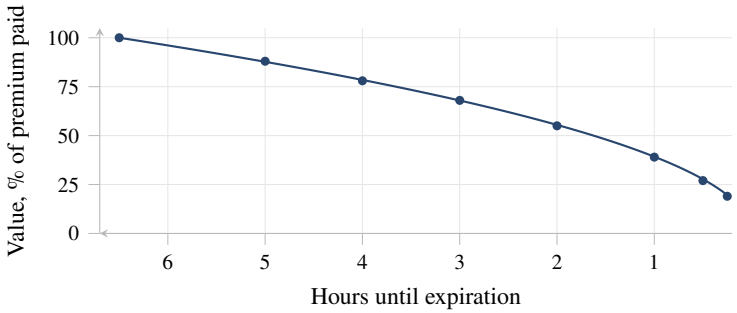


Figure 23.2: Value of an at-the-money 0DTE call through the session, with the index unchanged. Time runs left to right.

Example 23.1 What the buyer faces

An at-the-money 0DTE call on the index at 5000, bought at the open, costs 19.10, or **0.38% of spot**. A one-standard-deviation daily move is 47 points, or 0.94%.

The table gives the option's value through the day if the index does not move at all, and the rise the index needs for the buyer to break even on selling at each point:

Time left	Value, index unchanged	% of premium paid	Break-even rise
6.5 hours	19.10	100%	—
4 hours	14.94	78%	+0.15%
2 hours	10.53	55%	+0.28%
1 hour	7.43	39%	+0.35%
15 minutes	3.71	19%	+0.38%
Expiration	0	0%	+0.38%

An earlier exit needs a smaller rise, because the option is sold with time value still in it: with four hours left, 11.47 of the 19.10 recovered is time value.

Held to expiration, the buyer needs a 0.38% move in the right direction merely to get their money back, against a daily standard deviation of 0.94%. That is a hurdle of about 0.4 standard deviations before any profit. The option is not a bad bet in the way a lottery ticket is: it is priced

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without obvious error, with a negative expected return set mainly by the volatility risk premium of Section 19.10.

What makes it feel different is the *shape*. Sixty-one percent of the premium is gone by the last hour if nothing happens, so the position is either right quickly or worthless. The instrument converts a modest edge against the holder into a rapid and highly visible sequence of total wins and total losses, which is a design that human beings find much more engaging than a slow loss. Section 23.13 returns to that.

Note also what this does to bid–ask cost. At 19.10 a one-tick spread of ten cents (SPX's minimum tick for an option priced at $3 or more) is 0.52% of the premium; on the same option with fifteen minutes left, priced at 3.71, the same dime is 2.7%. Trading the same contract later in the day costs several times as much in percentage terms, and the volume is heaviest late.

23.4 CHARM: THE GREEK THAT ONLY MATTERS HERE

Section 18.6 named charm, the rate at which delta changes with the passage of time, and moved on, because for a monthly option it is a rounding error. Writing f for the option's value,

$$\text{charm} = \frac{\partial \Delta}{\partial t} = \frac{\partial^2 f}{\partial S \, \partial t}$$

with t the time elapsed. Charm is the cross-derivative that pairs with gamma: gamma is how the hedge changes when the price moves, charm is how it changes when the clock moves and the price does not. It is a cross term in $\Delta S \, \Delta t$ that Chapter 18's expansion, equation (18.1), leaves out, because for a monthly option the delta barely drifts with the clock alone. For 0DTE it is the dominant flow.

The reason is visible in Figure 23.1. As expiration approaches, an out-of-the-money option's delta must go to zero and an in-the-money option's to one. The whole delta surface is collapsing into a step, and that collapse happens on the clock rather than on the price. Figure 23.3 shows it on a single contract: with the index unchanged from open to close, the delta of a call struck just above spot falls the whole way down, and every point of that fall is a hedge somebody has to trade.

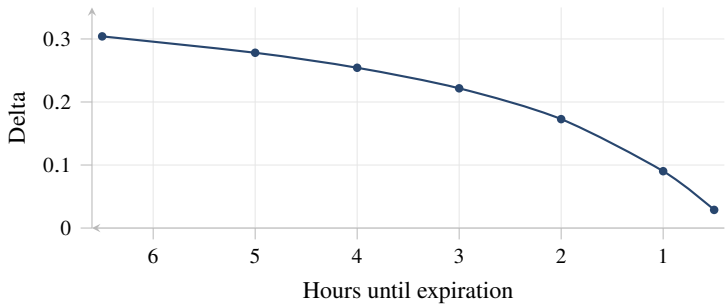


Figure 23.3: Delta of a call struck 0.5% above spot, with the index unchanged all day. Time runs left to right.

Example 23.2 A billion dollars of hedging, with the index unmoved

Take a call struck 0.5% above spot (about half a daily standard deviation, an ordinary strike for a 0DTE trade). With the index held at 5000 all day:

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Time left	Delta	Charm, per hour
6.5 hours	0.3041	−0.0146
4 hours	0.2543	−0.0274
2 hours	0.1728	−0.0614
1 hour	0.0902	−0.1099
0.5 hours	0.0288	−0.1254

A dealer short 10,000 of these contracts (index multiplier 100) hedges by holding 304,095 index units long at the open ($1.52 billion) and 90,164 units with one hour left, or $451 million. The dealer must *sell* roughly $1.07 billion of index by the last hour, with the index unmoved. Three properties distinguish this flow from the gamma hedging of Section 18.12.

It is one-directional. Gamma hedging buys on the way up and sells on the way down, netting out over a round trip. Charm hedging runs one way all day, set by the clock.

It is predictable. Anyone who knows the open interest by strike can compute it in advance, because the only input is the passage of time.

And it accelerates. The size of charm rises from 0.015 to 0.125 per hour, more than eightfold, so the flow is fastest in the final hour, which is also when 0DTE volume is heaviest and the day's other closing trades are competing for the same buyers and sellers.

Whether this stabilizes or destabilizes depends on the sign of dealer positioning, which Section 23.7 takes up. What is certain is that a large, predictable, clock-driven flow now exists in the final hour of the US equity session, and that ten years ago it was a fraction of its size.

23.5 THE OVERNIGHT GAP IS NOT IN THE CONTRACT

A quiet measurement problem sits underneath every comparison of 0DTE implied volatility with anything else, and it accounts for a discrepancy that is easily misread. A 0DTE option bought at the open expires at the close. It covers the session and not the night, yet it is quoted with a full trading day as its time to expiration. If the overnight period (close to next open) carries a fraction ω of daily variance, then the session carries $1 - \omega$, and the implied volatility that reproduces the option's price is

$$\sigma_{\text{session}} = \sigma_{\text{daily}} \sqrt{1 - \omega}$$

where σ_{daily} is the annualized volatility of close-to-close returns, the figure a 30-day implied volatility measures.

Overnight share ω	Implied volatility shown	Against 15% close-to-close
0%	15.00%	—
10%	14.23%	−5.1%
20%	13.42%	−10.6%
30%	12.55%	−16.3%
40%	11.62%	−22.5%

For a US index, where most news arrives during the session, ω is typically in the 15–25% range, so on the trading-day clock used here 0DTE implied volatilities print about a tenth below the 30-day implied volatility for purely mechanical reasons. (A calendar-minute clock makes the opposite error, showing 0DTE volatility well above the daily figure.) Cboe's one-day volatility index, VIX1D, launched in April 2023, publishes this session-only volatility, and comparing it with the 30-day VIX needs the same adjustment.

The lower print can be misread as evidence that short-dated options are cheap. They are not; the measurement window is shorter than the quoted maturity. Section 19.6's term structure is a real phenomenon, and this is not part of it; mixing the two produces a term structure with a spurious kink at the front.

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The effect has a practical corollary that matters more than the misreading. A 0DTE option cannot hedge an overnight gap, because the gap happens after it has expired. An institution using daily options to hedge event risk must hold a contract expiring after the event, and for an announcement made after the close that means tomorrow's contract, not today's.

23.6 THE SMILE AT ZERO MATURITY

Section 21.12 said that this chapter's market lives entirely in the region where classical stochastic volatility cannot produce a skew, and Section 21.7 gave the empirical law: the at-the-money skew grows like $T^{-0.4}$ as maturity shortens, measured from about a month to a few years. Whether it continues below a month is disputed; rough volatility (Section 21.8) says it does. Extrapolated to this market, the law is dramatic.

Maturity	Skew, relative to 30 days	Volatility points per 1% of moneyness
30 days	1.00	2.5
7 days	1.79	4.5
1 day	3.90	9.7
1 hour	8.24	20.6

A one-day smile is roughly four times as steep as the 30-day smile, and an hour before expiration it is eight times as steep. The ratios are the law's; the 30-day level of 2.5 points is a stylized anchor, several times the one-month slope of Table 19.3's surface (about 0.4 points per 1%), and every slope below scales with it. The slope is negative, so it is the puts that carry it: a put one percent below spot can show an implied volatility twenty points above the at-the-money level. The same slope cannot be run the other way for a call at the same distance (twenty points below a 15% at-the-money volatility is not a volatility), so the call wing flattens and turns up instead, and the smile at this horizon is a steep put wing against a shallow, convex call wing. Figure 23.4 draws the four maturities on one axis: from a month to an hour the put wing goes from a gentle slope to a nearly vertical one, while the call wing moves by a few points.

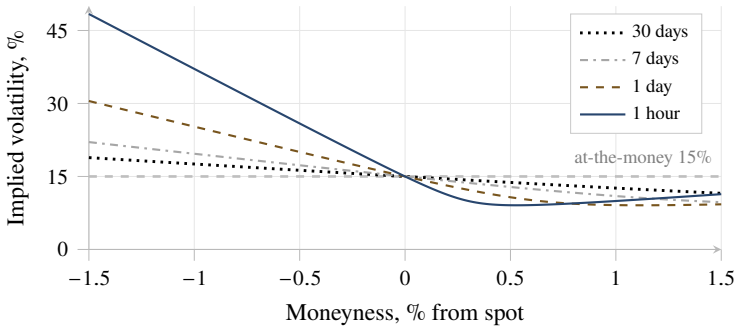


Figure 23.4: Implied volatility by moneyness at four maturities. Schematic: the at-the-money slope follows the $T^{-0.4}$ law of the table, extrapolated below one month, the at-the-money level is held at 15%, and one stylized wing shape is used throughout.

Three consequences follow, and the third is the one that matters for market structure.

The Black–Scholes implied volatility becomes a poor coordinate. Sections 15.7 and 19.12 described implied volatility as a language for quoting prices. At these maturities the language distorts: small differences in price map to enormous differences in implied volatility, and quoting in volatility terms conveys less information than quoting the price.

Only a model with jumps, roughness or very fast mean reversion can fit it. Example 21.3 showed jumps producing this shape (a smile whose range across strikes is 12.7 volatility

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points at one month and 1.5 at one year, almost all of it in the put wing), and Section 21.8 showed rough volatility producing the exponent directly. Whatever generates the 0DTE smile is not a classical diffusion with smoothly varying volatility, and this is the sharpest available evidence for that claim, because at these horizons such a path cannot travel far enough to give an out-of-the-money option any value; it takes jumps, very rough volatility, or a factor mean-reverting fast enough to act like either (Section 21.8).

And the steepness is where the retail flow is. The out-of-the-money puts that an intraday buyer of protection, or of a bet on a fall, naturally reaches for are precisely the strikes whose implied volatilities are most extreme. A buyer paying twenty points of skew is not making an obviously bad trade, since the steepness reflects a real probability of a downward jump, but they are buying the most expensive part of the surface, measured in the units the market itself uses. The call buyer at the same distance is on the cheap side of the same surface, paying less than the at-the-money volatility.

23.7 DEALER POSITIONING AND THE FEEDBACK LOOP

Section 18.12 established the mechanism, and the gamma term of equation (18.1) is its source: a move ΔS changes the dealer's delta by $\Gamma\,\Delta S$, and rehedging it trades with the move when dealers are *short* gamma, amplifying it, and against it when they are *long*, damping it. Applying that to 0DTE requires knowing which side dealers are on, and here the market's structure matters more than any model.

Retail buys options. The retail flow that Section 10.12 described is predominantly premium-paying: buying calls and puts outright. Against that flow dealers are short options and short gamma, and the feedback is amplifying.

But institutions sell them. A large and growing share of 0DTE volume is systematic premium selling—put spreads, call spreads, and iron condors (Section 12.3: a put spread below spot and a call spread above it, sold together) sold by funds harvesting the premium of Section 19.10 at the shortest available horizon. Against that flow dealers are long options and long gamma, and the feedback is dampening.

The two roughly offset, and the net is small and changes sign. This is the single most important empirical fact about the market and the one most often ignored. Studies of 0DTE order flow (Dim, Eraker, and Vilkov among them; see Further Reading) generally find it close to **two-sided**: customers buy options from dealers and sell options to dealers in roughly equal amounts, so dealers' net position is the difference of two large gross flows, and net dealer gamma from 0DTE contracts comes out a small fraction of that from longer-dated positions.

Example 23.3 How large the hedging flow is

Desks state gamma as the notional of the index that must be traded to rehedge a 1% move. A 1% move in the index changes the delta of one option by $0.01\,S\,\Gamma$ index units, so a position of n contracts with multiplier m needs

$$\text{hedge notional per } 1\% = 0.01\,n\,m\,\Gamma S^2$$

of index traded to restore the hedge. This is a different number from the gamma term of Table 23.1, the profit or loss on the same move, $\frac{1}{2}\Gamma S^2 \times 0.01^2$ per unit; the two describe one gamma, one as a trade and one as a P&L.

Suppose the dealer community is net short \$500 million of gamma per 1% from 0DTE positions. From Table 23.1 that is the gamma of about 930 at-the-money contracts an hour before expiration ($0.01 \times 100 \times 0.021528 \times 5000^2$ = \$538,200 each, the 100 being the multiplier m), or 13,000 at thirty days. Being short gamma, the dealers gain delta as the index falls and lose it as the index rises, so they must sell futures into a fall and buy into a rise; to first order, holding gamma fixed, the amount is \$500 million for each 1%:

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Index move	Index futures dealers must sell (fall) or buy (rise)
0.5%	$250m
1.0%	$500m
2.0%	$1,000m
3.0%	$1,500m

Each trade follows the move that caused it, which makes the flow amplifying. Gamma does not stay fixed at this horizon (Section 23.10): for the one-hour book the true figures are about $190m on a 0.5% move and stop near $230m, the whole of the book's delta, beyond 1%; for the thirty-day book the table is close.

Against S&P futures volume of several hundred billion dollars notional a day, these are small numbers in aggregate. A billion dollars of futures is minutes of normal volume.

They are not small at 3:55 p.m. New York time. The flow is concentrated in the final hour, by both charm and gamma, and within that window it can all be on one side. The close is the busiest part of the futures day, but a hedger who must sell now does not get to trade against the day's volume. They trade against what other participants are willing to buy at that moment at the best bid, which is a few hundred E-mini S&P 500 futures contracts (each $50 times the index, about $250,000 at 5000), or tens of millions to about a hundred million dollars. That is the right comparison, not daily volume, and against it a few hundred million dollars that all wants the same side is material.

23.8 JUMPS WITHIN THE DAY

Section 21.5 observed that jumps cluster across underlyings (the events that make one stock gap tend to make others gap) and left the intraday side to this chapter. For 0DTE the scheduled events behind that clustering are not a refinement—they are the dominant term.

Volatility within a session is not uniform. It concentrates at the open, at scheduled data releases, at policy announcements, and into the close. For a monthly option this averages away; for an option with six hours to live it does not. A 0DTE implied volatility is therefore not a number but a *schedule*, and a public one.

Example 23.4 What a two o'clock announcement does to a morning option

Take Section 23.5's session: 15% close-to-close volatility, 20% of variance overnight, so an ordinary session's standard deviation is 0.845% and a 0DTE option shows 13.4% implied volatility.

Assume ordinary variance is spread evenly through the session. Now add a scheduled announcement whose move has standard deviation s_e. Variances add, so the session's standard deviation becomes $\sqrt{0.845\%^2 + s_e^2}$; the last column is the implied volatility as a multiple of the 13.4% with no event:

Event move (s.d.)	Session s.d.	Implied volatility shown	Multiple
none	0.845%	13.4%	—
0.50%	0.982%	15.6%	1.16
0.75%	1.130%	17.9%	1.34
1.00%	1.309%	20.8%	1.55
1.50%	1.722%	27.3%	2.04

A single scheduled event with a one-percent standard deviation raises the morning's 0DTE implied volatility by more than half. Nothing about the market's ordinary volatility has changed; one known future minute has been added to the measurement window.

The sharper effect is what happens when the minute passes. Compare the option at 1:59 p.m. and at 2:01 p.m., with two hours left, the index unchanged, and the 1.00% event of the table:

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	Remaining s.d.	Annualized
Just before the announcement	1.104%	31.6%
Just after	0.469%	13.4%

The implied volatility falls by a factor of 2.36 in two minutes. The event variance has simply left the window. Three points follow.

This is not a crash in volatility; it is the calendar. A trader who buys the option at 1:59 and sells it at 2:01 without a view on the announcement has bought the event and sold the aftermath, which is a coherent trade. A trader who buys at 1:59 hoping to profit from a rise in volatility has misunderstood the instrument.

The calendar makes the constant-volatility assumption untenable, and not as a modeling nicety. Section 15.1 listed constant volatility among the Black–Scholes assumptions and Chapter 21 relaxed it. At this horizon the failure is not that volatility is stochastic—it is that volatility is *scheduled*, and everyone has the schedule. Desks price 0DTE against a calendar of event weights rather than a single number, which is a different kind of model from anything in Chapter 21; Section 23.9 describes it.

Finally, the clustering that Section 21.5 attributed to jumps is visible here directly. The events that make one index gap are the same events (data, policy, geopolitics) that make others gap, so intraday jump risk is correlated across underlyings exactly when it is largest. A book short out-of-the-money 0DTE options on several indices is not diversified; it is one position.

23.9 PRICING A 0DTE OPTION

Every number so far has come from the Black–Scholes–Merton formula with a 15% volatility and a clock in trading hours, and for the Greeks that was the right tool: the shape of gamma, charm and decay does not depend on the details. A reader who has followed the chapter this far will have two questions the formula does not answer. Which options are actually being traded? And if constant volatility is untenable, what does a desk use instead?

Which strikes trade. Mostly they are not the at-the-money ones. Cboe's own tabulation of 2023 volume and the academic samples named in the Further Reading agree on the shape. The typical contract is near the money but out of it (roughly a quarter to one percent away, a delta between about 0.10 and 0.35), and puts and calls trade in roughly equal numbers, the puts reaching further from spot than the calls do. Far out-of-the-money strikes, one to two percent away on a quiet day, account for many contracts and very little premium, because they cost a few dollars at most, and cents by the afternoon; counted by premium rather than by contract, the center of the market moves closer to the money. The distribution also moves through the day: in the morning the traded strikes span about a percent either side of spot, and by the last hour nearly everything is within about half a percent, the one-hour band of Figure 23.1, because that is the only place where any time value is left. In-the-money options trade little. By put–call parity, an in-the-money call is the out-of-the-money put at the same strike plus a long forward position: its time value is the put's, and the rest is index exposure bought at an option's wider bid–ask spread. That time value drains away through the day: at the open a call half a percent in the money carries about 15 points of it, an hour before the close about 4, and with fifteen minutes left about 1, at the skewed volatility of Section 23.6. A trader who wants the exposure buys the futures contract, and one who wants the option buys the out-of-the-money side. Retail flow is mostly single-leg, an outright call or put opened and often closed within the session, while institutional flow is mostly spreads: a put spread half to one percent below spot, a call spread the same distance above, or both together as an iron condor, held to expiration. Section 23.13 says who these people are; the point here is that the market's pricing problem is concentrated in a narrow band of near-the-money strikes and a steep wing of puts just below it.

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Why one volatility will not do. The Black–Scholes–Merton model makes three assumptions that are tolerable for a monthly option and fail together at this horizon. Volatility is constant, so variance accrues evenly with the clock; but Section 23.5 showed that a fifth of the day's variance falls in a window the contract does not cover, and Example 23.4 showed a scheduled minute carrying several times as much variance as the rest of the afternoon. The price path is continuous, so a strike several standard deviations away is nearly worthless; but at one hour a strike one percent below spot is 2.7 standard deviations away, and Section 23.6 showed the market pricing it as though the distance were far smaller, because a jump can cover it in an instant. And the hedge is continuous and free, so the price is the cost of replication; but Table 23.1 put gamma at fourteen times its monthly value and Example 23.1 put a dime of bid–ask at 2.7% of the premium, so the replication cost is neither small nor well defined. Example 23.5 shows how large the resulting error is.

Example 23.5 What a flat volatility gets wrong an hour before the close

Price a put below spot in the chapter's setting (index 5000, 4% rate, 1.5% dividend yield), first with a flat 15% volatility and then with the volatility given by the at-the-money slope of Section 23.6, extended linearly: 15% plus 2.5 points per percent of moneyness at thirty days, and plus 20.6 points per percent at one hour.

Time left	Strike	Flat 15%	With the skew	Ratio
30 days	0.5% below spot	83.99	92.45 (at 16.2%)	1.1
30 days	1.0% below spot	73.30	89.90 (at 17.5%)	1.2
1 hour	0.5% below spot	0.75	3.72 (at 25.3%)	5.0
1 hour	1.0% below spot	0.02	2.75 (at 35.6%)	148

At thirty days the smile is a correction of ten or twenty percent to a price the flat model gets roughly right. At one hour it is the price. The flat model values the put one percent below spot at two cents, because under a continuous path with a 0.37% hourly standard deviation the index almost never gets there; the market values it at about 2.75, over a hundred times as much, because a jump can. A desk that priced this put with the at-the-money volatility would offer it at the minimum tick and hedge it as if it were worthless.

What desks use instead. There is no one replacement; practice is closer to Chapter 19's than to Chapter 21's: the surface is the model, with the pieces assembled around it.

The first piece is the clock. Time to expiration is measured not in minutes but in expected variance remaining: an intraday profile heavy at the open and into the close with a thin middle, plus the weights of the day's scheduled events. This is **business time**, and the Black–Scholes–Merton formula is used with $\sigma^2 T$ replaced by that variance budget. It is why the option of Example 23.4 lost more than half its implied volatility in two minutes with nothing else changing: the budget was spent, not the volatility revised.

The second is the forward, which comes from the E-mini futures contract, the instrument the position will be hedged with, or from put–call parity within the chain as the VIX construction of Section 22.6 does. The smile is quoted against that forward, not against the cash index.

The third is the smile itself, fitted expiration by expiration in the standardized moneyness of Section 19.1,

$$\frac{\ln(K/F)}{\sigma\sqrt{T}},$$

the strike's distance from the forward in standard deviations, with σ the at-the-money implied volatility and T the business time remaining. The form is low-parameter and arbitrage-free by construction (the SVI parameterization of Section 19.5 is common), refitted through the day. In raw moneyness the smile steepens sharply as expiration approaches; in standardized

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coordinates most of that steepening disappears, and what is left, a slope proportional to $T^{0.1}$ (the $T^{-0.4}$ times the $\sqrt{T}$ the standardization divides by), is the residual the fit has to carry. The wings are where a view on jumps enters: a desk that expects an event to be larger than the calendar weight implies lifts the put wing directly, rather than changing a parameter in a dynamic model. The models of Chapter 21 are used off the desk, to check that the fitted surface is one some process could have produced. Jump models give a short-end skew that overshoots the market's, as Section 21.7 found, and rough volatility gives the $T^{-0.4}$ the market shows, but they do not set the quote.

The last piece is inventory. The spread and the skew adjustment around the fitted midpoint are set by the dealer's own position and by customers' orders: quotes are shaded against the book's gamma and charm strike by strike, widen into the close, and move with the flow. What the model supplies is the hedge ratios that make that possible, and for an actively traded 0DTE strike that is its whole job. The price is set by the bids and offers already in the market, not by anyone's formula.

23.10 PINNING, EVERY DAY

Section 10.9 introduced **pinning** alongside pin risk at the monthly expiration: the underlying is drawn toward a strike with large open interest, because dealers hedging a long gamma position sell into rallies and buy into declines around it. With a daily expiration, that mechanism now runs every session. Example 23.6 computes the size of the restoring force, because it is far larger than the monthly intuition suggests, and Figure 23.5 plots it.

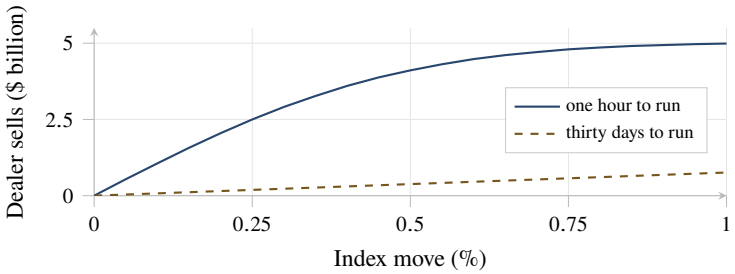


Figure 23.5: Dealer sale against index move for 20,000 calls held long at the money.

Example 23.6 The restoring force at one hour

Dealers are long 20,000 calls at one strike, the index sits at it, and one hour remains.

Index move	Delta goes from	Dealer sells
+0.10%	0.502 → 0.609	$1,063m
+0.25%	0.502 → 0.752	$2,499m
+0.50%	0.502 → 0.912	$4,114m
+1.00%	0.502 → 0.996	$4,990m

With thirty days to run the same book sells $382 million on a 0.5% move and $760 million on a 1% move, the straight line of Figure 23.5: doubling the move doubles the sale, which the one-hour curve plainly does not. Two features of the one-hour figures matter and they point in opposite directions.

Small moves meet an enormous restoring force. A tenth of a percent generates a billion dollars of stabilizing supply. This is genuine pinning, and it is why the index often trades in a narrow range near a strike with large open interest.

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But the force saturates. By a 1% move the delta has reached 0.996 and there is nothing left to sell: the last half-percent of the move generated only $876 million of additional hedging against $4.1 billion for the first half. The stabilizer is strongest for moves that did not need stabilizing and absent for the ones that did.

That asymmetry is the defensible form of the worry in Section 23.14. A market that suppresses small fluctuations and cannot suppress large ones does not merely fail to help in a crisis—it makes the crisis look more surprising, because the small moves that would ordinarily have signaled building stress were damped away.

23.11 MEASURING DEALER GAMMA, AND WHY THE NUMBER IS UNRELIABLE

Everything in Section 23.7 depends on the sign of net dealer gamma, and that quantity is not observable. The standard construction, which Section 18.12's case study introduced, takes the open interest OI_i of each listed contract i (a call or a put at some strike), computes its gamma Γ_i, and assigns the dealers a side, $s_i = +1$ (long) or -1 (short). The usual rule assumes them long all call open interest and short all put open interest. The published "gamma exposure," in the form SqueezeMetrics (2017) first set out, is then $\sum_i s_i \,\mathrm{OI}_i \, m \, \Gamma_i \, S^2 \times 0.01$, with m the contract multiplier (100 for SPX): the hedge notional per 1% of Example 23.3 (Section 18.7's gamma in dollars) with the signs assumed, and the "flip level" is the index level at which it changes sign.

The trouble is that the sign assumption is asserted, not measured. Commercial versions infer the side from trade data rather than assume it, which narrows the problem without removing it; Example 23.7 shows how much of the answer the assumption decides.

Example 23.7 The assumption is the answer

Take 100,000 calls and 100,000 puts outstanding at the same strike, with identical gamma per contract. Assume dealers are on one side of every contract. Let w_c be the fraction of the calls dealers are long (they are short the rest) and w_p the fraction of the puts they are short (they are long the rest), so that the standard rule is $w_c = w_p = 100\%$. Net dealer gamma is then 200,000 $(w_c - w_p)$ contracts' worth.

Assumed split (calls long / puts short)	Net dealer gamma (thousands of contracts)
100% / 0%	+200
80% / 20%	+120
60% / 40%	+40
50% / 50%	0
40% / 60%	−40
100% / 100%	0

The sign flips between the 60/40 and the 40/60 assumption, and nothing in public data determines which is right. The last row is the standard rule itself: applied to equal call and put open interest it returns exactly zero, so here the published number would be zero not because dealers are flat but because the assumption is symmetric. Open interest records that a contract exists; it does not record who is on which side.

A published gamma-exposure level is therefore what dealer positioning would be if the assumed split were true, a weaker statement than it usually receives. Three points follow:

The assumption is worst exactly where it matters. It is roughly defensible for a market dominated by call overwriting (investors selling calls against stock they hold) and institutional put buying. Section 23.7 argued that 0DTE flow is close to two-sided, which is the case in which the naive split is furthest from the truth.

The flip level inherits the error. A published "gamma flip" at some index level is a function of the assumed signs, so two analysts with different assumptions produce different flip levels from

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identical data.

None of this means the concept is useless. Dealer gamma is real, and Sections 18.12 and 23.10 depend on it; published estimates simply measure open interest precisely and positioning by assumption.

23.12 DOES 0DTE AMPLIFY OR DAMPEN?

This is the question the market cares about, and the answer is that the evidence is mixed and the sample is short.

The case that it amplifies. The mechanism of Section 23.7 is real. Gamma is enormous and concentrated in a narrow band, so hedging demand is a step function of the index level. Charm produces a large one-directional flow into the close. And the positions are concentrated and the hedging mechanical, which is the pattern that produced 1987 (Section 40.2), February 2018 (Section 22.8) and the snowball knock-ins of Section 35.7.

The case that it dampens. Short-dated option sellers, in aggregate, supply gamma to the market. A dealer long gamma sells into rallies and buys into declines, which is stabilizing. If systematic premium selling dominates, 0DTE makes the intraday market *less* volatile, not more.

What the data show. Realized intraday volatility on the S&P 500 has not risen in a way that tracks 0DTE growth (Dim, Eraker, and Vilkov 2024). There has been no episode in which 0DTE hedging has been convincingly identified as the cause of a large move, though two stress episodes have tested the market since 0DTE became its largest segment: August 5, 2024 and the April 2025 tariff sell-off, when VIX closed above 50. Accounts of both, the BIS's of August 5 among them, trace the moves to news and to leveraged positions being unwound, and on August 5 to wide pre-open option quotes that inflated the VIX itself, not to 0DTE hedging. And the aggregate net dealer gamma attributable to 0DTE appears small relative to the longer-dated book. Against this, Brogaard, Han, and Won find a rise in volatility and attribute it to speculative retail trading rather than to dealer hedging.

So the fair summary is: the mechanism exists and it has not yet produced an incident. Four reasons explain the absence. The flow is two-sided, the main reason and the most fragile. Expiration is a natural stop-loss, since tomorrow the position does not exist. Margin is collected daily and the positions are cleared: the 2018 episode ran through an exchange-traded note, XIV, whose prospectus allowed early termination (Section 22.8), while a listed option position is margined by the clearing house of Chapter 2 against a position that self-liquidates. And the underlying is the deepest equity market in the world. The last three are structural rather than behavioral, which, by the argument of Section 22.5's case study, is the kind that holds.

That summary deserves caution. *The stabilizing property is a property of the flow, not of the instrument.* Nothing guarantees that 0DTE order flow stays two-sided. It is two-sided because two groups happen to want opposite things at the same horizon, and either could stop.

Positions do not accumulate, which is genuinely reassuring. A 0DTE book is gone by the close. Compare Section 22.8's inverse products, whose short position was carried from day to day for years, or the snowball books of Chapter 35, whose knock-in exposure grew as the market fell over months. Whatever risk 0DTE creates is created and destroyed within one session, which puts a hard bound on how large it can get.

But the bound is on size, not on speed. A structure that resets daily cannot build a slow-motion catastrophe; it can still produce a violent hour. The events that would test this market (a large gap that arrives while dealers are short gamma near the close) are rare.

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Case Study 23.1 The same mechanism in a shallow market

The argument of Section 23.12 rests heavily on the depth of the S&P 500. What the identical mechanism does without that condition is documented in a 2021 SEC staff report.

In January 2021 a heavily shorted small-capitalization US stock attracted very large retail call buying, much of it in short-dated, far out-of-the-money strikes. The dealers who sold those calls were short gamma, and Section 23.7's arithmetic applied unchanged: as the stock rose, their delta grew, and they had to buy stock to stay hedged.

Everything is the same as the index case except one number. The S&P 500 absorbs a billion dollars of hedging in minutes. A stock with a small free float, a large fraction of which was already lent out to short sellers, could not absorb the equivalent at all. The hedging demand was large relative to the shares actually available, so it moved the price, which increased the hedging demand, which moved the price again. Three points need separating, since the episode is usually told as one about intent.

The dealers were not speculating. They were hedging a position they had sold, and a hedge that moves the thing it hedges is a structural feature of delta hedging in a market too small for it.

The short interest (shares sold short and not yet bought back) mattered as much as the options. Short sellers covering are buyers too, and the two flows reinforced one another. Attributing the whole move to option hedging overstates a mechanism that was real but not alone. That is the same caution Section 23.12 applies to 0DTE.

And the binding constraint was free float, not leverage. The options were not unusually leveraged; the underlying was unusually small, and that is the variable that decides whether the feedback of Section 23.7 is a curiosity or an event.

The lesson is therefore not that 0DTE is safe. The mechanism is settled, and the outcome depends on the ratio of hedging demand to available liquidity, which is small for the S&P 500 and was not for a single small-cap stock. To ask whether 0DTE is dangerous is to ask whether that ratio can change, and Section 23.14 lists the ways it could.

23.13 WHO TRADES 0DTE, AND WHY

Retail directional traders buy 0DTE options as a leveraged intraday bet. Example 23.1 gives the economics: a modest negative expected return, delivered as a rapid sequence of complete wins and complete losses.

Systematic premium sellers sell 0DTE spreads to harvest the volatility risk premium at the shortest horizon, where the annualized premium looks largest and, crucially, where a position cannot be carried into a multi-day drawdown. Selling a defined-risk spread that expires in hours is a different risk from selling a monthly straddle: the loss is bounded and the position cannot be held past its worst moment.

Institutional hedgers use 0DTE tactically around events within the session (a data release, a policy decision), and buy the next day's expiration for one announced after the close, where the risk is genuinely one-day and paying for a month of protection is waste. This is the least remarked and possibly the largest legitimate use, and it is a straightforward improvement in market completeness: a one-day risk can now be hedged with a one-day instrument.

Dealers and market makers intermediate, and their positioning is the transmission mechanism of Section 23.7.

The instrument itself is neutral: Section 12.8's replication argument, run across expiration dates instead of strikes, says a finer set of maturities strictly enlarges the space of achievable payoffs. What is new is the concentration of hedging it induces, which should be assessed

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separately.

23.14 WHAT WOULD HAVE TO GO WRONG

The useful form of the question is not "is 0DTE dangerous" but "what configuration would make it so," because that is checkable in advance.

One-sided flow. If the systematic sellers withdrew (after a bad quarter, or a capital charge, or a change in fund mandates), the retail buying would leave dealers uniformly short gamma.

A gap into concentrated gamma. The dangerous configuration is a large move arriving in the last hour, when the gamma band of Figure 23.1 is at its narrowest and the charm flow of Example 23.2 is at its fastest. A 1% move then takes the index outside the band, and the hedging that would have absorbed it runs out partway through, as Example 23.6 showed.

Liquidity that is not there. All of the numbers in Example 23.3 are small against average volume and large against the depth available in a fast market. The mechanism that matters is the one Section 2.9 identified in margin procyclicality: in a fast market the participants who would provide liquidity are the same ones who suddenly need it. And short-dated hedging is reaching shallower markets: since January 2026 stocks with a market value above $700 billion also carry Monday and Wednesday expirations (Section 10.2), which is the setting of Section 23.12's case study rather than the index's.

None of this is a prediction. It is a specification of what to watch, and the watchable quantities (the open-interest distribution near spot, the share of volume that is premium-selling, and estimates of dealer gamma by strike, read with Section 23.11's caution) are published or computable. The risk is not hidden, and unlike the episodes of Sections 22.8, 35.7 and 40.2, the profession is watching this one while it is small.

SUMMARY

Same-day options went from a single-digit share of SPX option volume to more than half in under a decade. Volume overstates the risk transferred; it does not overstate the hedging, which is compressed into hours.

Everything peculiar follows from the Greeks as $T \to 0$. Gamma an hour from expiration is many times gamma at thirty days, and the gamma band collapses to a narrow range around spot, so hedging demand becomes a step function of the index level. Charm becomes the dominant flow: a dealer short calls struck just above spot unwinds a large hedge in a day the index does not move, one way, predictably, fastest in the final hour. The buyer of an at-the-money call pays well under one standard deviation of the day's move and has lost most of the premium by the last hour if nothing happens.

Three things are routinely misread. A 0DTE option covers the session but is priced with a full day, so it shows less implied volatility than the daily figure and cannot hedge an overnight gap. Extrapolated below a month, where the evidence is disputed, the $T^{-0.4}$ law steepens the skew by nearly an order of magnitude from thirty days to the last hour before expiration. And volatility within the day is scheduled: an announcement lifts the morning's implied volatility, and the remaining option's falls sharply when it passes. One volatility therefore cannot price the market: a put just below spot an hour before the close is worth almost nothing at a flat volatility and many times as much with the skew—so desks price on a business-time clock against a smile fitted in standardized moneyness, and the model supplies hedge ratios, not the price.

Whether hedging amplifies or dampens depends on the sign of dealer gamma, which is not observable: retail buying and systematic selling roughly offset, and published estimates measure open interest precisely and positioning by assumption. Pinning happens daily and saturates: the hedging flow is heaviest on a small move and spent by a larger one. There has been no incident because the flow runs in both directions, positions expire before they

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accumulate, margin is daily, and the underlying is deep; the first of these is the one most likely to change, and the configuration to watch is flow running one way meeting a large move in the final hour.

FURTHER READING

This is the newest topic in the book and the literature is accordingly unsettled; a reader should expect the empirical picture to have moved. Cboe's own research notes on 0DTE volume composition (2023) are the source for what is actually traded and by whom, and they separate contract counts from risk transfer, which most commentary does not. On order flow and dealer positioning, Brogaard, Han, and Won (2023, revised 2026) find that 0DTE trading raises the volatility of the underlying and attribute the effect mainly to speculative retail trading rather than to dealer hedging, Dim, Eraker, and Vilkov (2024) find market makers' net 0DTE gamma small and, on average, long, with calmer rather than wilder trading after it, and Beckmeyer, Branger, and Gayda (2023) document the retail side (who buys, at what strikes, and with what returns); together they are the evidence Section 23.12 calls mixed, and they disagree less about dealer hedging than about whether 0DTE trading moves volatility at all. Bandi, Fusari, and Renò (2023) is the pricing paper, with time-varying skewness and kurtosis and jumps for the wings. On the feedback mechanism itself, the literature on option hedging and market impact predates 0DTE by decades and applies unchanged, and Grossman (1988) and the Brady Commission report are the place to start.

PRACTICE QUESTIONS

23.1 Explain why 0DTE contract volume overstates the amount of risk being transferred.

23.2 State how at-the-money gamma behaves as $T \to 0$, give the ratio between one-hour and thirty-day gamma from Table 23.1, and explain why the narrowing of the gamma band matters more than the growth in gamma itself.

23.3 An at-the-money 0DTE call costs 0.38% of spot. State the move required to break even at expiration, compare it with a one-standard-deviation daily move, and explain why the bid–ask spread is a much larger percentage cost late in the day than early.

23.4 Define charm, explain why it is negligible for a monthly option and dominant for a 0DTE option, and use Example 23.2 to state three properties that distinguish charm hedging from gamma hedging.

23.5 The overnight period carries 25% of daily variance. Compute the implied volatility a 0DTE option will show in a market whose close-to-close volatility is 16%, explain why this is not a term-structure effect, and state which contract would hedge an announcement made after the close.

23.6 Using the $T^{-0.4}$ law, compute how much steeper a four-hour smile is than a 30-day smile, and explain why quoting a 15-minute option in implied volatility terms conveys less information than quoting its price.

23.7 Explain why the pinning force of Example 23.6 saturates, and why a market that damps small moves but not large ones may be worse than one that damps neither.

23.8 A published gamma-exposure figure assumes dealers are long all calls and short all puts. State what is measured and what is assumed, and explain when the assumption is least defensible.

23.9 A 0DTE option shows 13.4% implied volatility with no event scheduled. A 2 p.m. announcement adds a move of 0.8% standard deviation. Compute the morning implied

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volatility and its value just after 2 p.m., with two hours left, and explain why the fall is not a fall in volatility.

23.10 Explain why retail flow makes dealers short gamma and systematic premium selling makes them long. A dealer book is net short \$800m of gamma per 1%. Compute the hedging trade a 1.5% move requires, state its direction, and explain why the same flow can be negligible against daily volume and material against the depth at 3:55 p.m.

23.11 State the four reasons Section 23.12 gives for the absence of an incident, identify which is behavioral rather than structural, and explain why daily expiration bounds the size of the risk but not its speed.

23.12 Explain what happens to dealer hedging demand when a move takes the index outside the gamma band, and why this is the dangerous configuration.

FURTHER QUESTIONS

23.13 Section 23.2 reports the gamma band's upper edge closing in from +6.76% to +0.44%. Derive the band's edges as a function of σ and T, show why the band is symmetric in log-strike about a point slightly above spot rather than about spot, verify the three pairs of figures, and determine how the band would change if implied volatility doubled. Discuss what this implies about whether 0DTE is more or less dangerous in a high-volatility regime.

23.14 Example 23.2 finds a \$1.07bn hedge unwind with the index unmoved. Derive the charm flow between the open and the last half hour for a book with a given open-interest distribution across strikes, explain why over that window the flow is largest for strikes roughly half a daily standard deviation from spot rather than at the money, say where the at-the-money strike's flow goes instead, and discuss what this implies about where in the strike distribution to look for the largest predictable flow.

23.15 Reproduce the one-hour rows of Example 23.5, then price the same two puts in Merton's jump-diffusion of Section 21.5 with the diffusion volatility set to 15%, and find the jump intensity and mean jump size that reproduce the skewed prices. Comment on how many parameter pairs do so, and on what that says about calibrating a dynamic model to a single expiration.

23.16 Section 23.12 argues the market is benign mainly because the flow is two-sided, and that this is the fragile reason. Construct a model of the two groups (retail buyers and systematic sellers) in which the equilibrium is stable, then identify the parameter changes that would break it, and state what observable series would give warning.

23.17 Section 23.14 claims that the hedging that would absorb a move "runs out partway through" once the move takes the index outside the gamma band. Formalize this: write dealer hedging demand as a function of the index level for a book concentrated at one strike, show where it is increasing and where it vanishes, and assess whether a mechanism that absorbs small moves and not large ones is worse than one that absorbs neither.

23.18 Section 23.13 argues that a daily expiration is a genuine improvement in market completeness. Make the argument precise using Section 12.8's replication result, then construct the counter-argument that finer maturities can reduce welfare by inducing hedging concentration, and state what would have to be measured to decide between them.

23.19 Compare 0DTE with portfolio insurance (Section 40.2), snowball autocallables (Section 35.7) and inverse volatility products (Section 22.8) along the dimensions of crowding, mechanical hedging, accumulation, and observability. Explain which features 0DTE

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shares and which it does not, and use the comparison to argue either that the profession's current level of concern is about right or that it is misdirected.

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